On Heterogeneity and Edgeworth Complementarity Between Public and Private Consumption: Implications for the Fiscal Multiplier

Preliminary and incomplete – please do not circulate

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Abstract

Does it matter how households value publicly provided goods relative to private consumption, and does this depend on where they sit in the income distribution? We show that it does – and that both questions have first-order implications for the size of the fiscal multiplier. Embedding Edgeworth complementarity between public and private consumption into a tractable heterogeneous-agent New Keynesian (THANK) model, we derive a closed-form fiscal multiplier and characterize two new channels absent from standard frameworks. The first is a *direct fiscal demand* channel through which government spending stimulates private consumption above and beyond standard income effects. The second emerges when poorer constrained households treat public goods as complements while wealthier households substitute privately – a configuration strongly supported by evidence that public education, healthcare, and transport disproportionately benefit low-income households. This distributional asymmetry generates curvature heterogeneity across household types that *decouples income cyclicality from the hand-to-mouth share*, providing a new route to the MPC/MPE trilemma that operates through the government-spending margin rather than through labor supply frictions. A distributional dominance condition governs the aggregate multiplier: fiscal policy is most effective precisely in the economies where it is most needed – those with high inequality and strong complementarity between public services and the private consumption of the poor. Taking the model to a quantitative HANK calibration disciplined by US micro data, we show that plausibly heterogeneous internalization raises impact multipliers by 0.12 to 0.34 log points relative to the separable benchmark, with welfare gains from targeted fiscal expansion of up to 1.8% of steady-state consumption for the bottom quintile. The framework provides a transparent mapping from the distributional incidence of public goods to aggregate fiscal transmission, with direct implications for developing-economy policy design.

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1 Introduction

How large is the fiscal multiplier, and what determines its size? A substantial empirical and theoretical literature has identified three facts that any serious model of fiscal transmission must confront simultaneously [Auclert et al., 2024a]:

Fact 1. *Marginal propensities to consume (MPCs) are high.* Households spend approximately 25 cents of every additional dollar of income in the quarter they receive it, and around 50 cents annually [Kaplan and Violante, 2014, Fagereng et al., 2021].

Fact 2. *Marginal propensities to earn (MPEs) are low.* When households receive a windfall, they reduce their labor supply very little; empirical estimates place the annual MPE between 0 and 0.04 [Cesarini et al., 2017].

Fact 3. *Fiscal multipliers are moderate.* Under accommodative monetary policy, both impact and cumulative multipliers lie in the range 0.6–2.0 [Ramey, 2019, Christiano et al., 2011, Woodford, 2011].

Auclert et al. [2024a] show that these three facts create a *trilemma* for New Keynesian models with frictionless labor supply: it is impossible to match all three simultaneously with standard separable preferences. Matching Facts 1 and 2 requires strong consumption–labor complementarity ($U_{cn} \neq 0$), but this same complementarity generates multipliers that violate Fact 3. Bilbiie et al. [2025] resolve the trilemma through a different non-separability: by separately parameterizing complementarity and income effects in $U(c, n)$, they allow the model to target MPCs and MPEs independently of the multiplier.

This paper introduces a *new dimension* to this debate. Both Auclert et al. [2024a] and Bilbiie et al. [2025] work with utility over consumption and labor: $U(c, n)$. We extend the utility function to $U(c, g, n)$, where g is government spending – a publicly provided good that directly enters household preferences. The cross-partial $U^j_{cg} \neq 0$ is *Edgeworth complementarity* between private and public consumption: a rise in g raises (if $U^j_{cg} > 0$) or lowers (if $U^j_{cg} < 0$) the marginal utility of private consumption. This is empirically well-documented [Karras, 1994, Jalles and Karras, 2022, Francois and Keinsley, 2023, Francois, 2023, Barthel and Francois, 2025] and generates a fiscal-transmission mechanism that is entirely absent from models where government spending enters only through the resource constraint.

Contribution. Our central contribution is threefold, with each point corresponding to an empirically-testable new channel:

1. We characterize a **direct fiscal channel** ξ – absent from the separable framework of Bilbiie [2025] – that captures how government spending directly stimulates private demand through its effect on the marginal utility of consumption. This channel depends on the cross-elasticities

ϵ^j_{cg} and is new to the literature. Crucially, ξ and the standard income-cyclicality statistic χ are cleanly separated: χ is identified by ϵ^j_{cc} moments alone, while ξ is identified by ϵ^j_{cg} moments – giving researchers two independent sufficient statistics for fiscal transmission.

2. We show that **heterogeneous internalization** ($\theta^H \neq \theta^S$) generates curvature heterogeneity across household types. Since $\epsilon^j_{cc} = \sigma\alpha^j$ under the linear specification, $\theta^H \neq \theta^S$ induces $\alpha^H \neq \alpha^S$, which shifts income cyclicality χ *independently* of the hand-to-mouth share λ. This θ-induced curvature heterogeneity provides a new route to the MPC/MPE trilemma resolution of Auclert et al. [2024a] – one that operates through the government spending margin rather than through consumption–labor non-separability, and works even when $U_{cn} = 0$ exactly.

3. We derive a **distributional dominance condition** linking inequality, complementarity, and multiplier size: fiscal policy is most effective precisely where it is most needed – in unequal economies where the poor disproportionately complement public goods with private consumption. Quantitatively, in a HANK calibration disciplined by US micro data, moving from the separable benchmark to heterogeneous internalization raises impact multipliers by 0.12–0.34 log points.

The direct fiscal channel: a new sufficient statistic. The first contribution deserves emphasis because it stands independently of the trilemma discussion and is arguably the paper's cleanest result. In Bilbiie [2025]'s separable framework, the fiscal multiplier depends on one sufficient statistic: income cyclicality χ. Government spending affects output only through general-equilibrium income effects – it raises wages and profits, which raise hand-to-mouth income, which feeds back into demand. There is no channel by which g directly affects the private consumption decision.

Introducing $U^j_{cg} \neq 0$ creates a second sufficient statistic: ξ, the direct fiscal channel. When g rises, the marginal utility of private consumption changes – upward for households with $\epsilon^j_{cg} < 0$ (complements) and downward for those with $\epsilon^j_{cg} > 0$ (substitutes). This directly alters how much hand-to-mouth households spend out of any given income, above and beyond the standard wealth and income effects. Formally, ξ captures the direct response of hand-to-mouth consumption to the fiscal shock: $\hat{c}^H_t = \chi\,\hat{c}_t + \xi\,\hat{g}_t$, where the $\xi\,\hat{g}_t$ term is entirely new. In Bilbiie [2025]'s separable world, $\xi = \zeta(\chi - \mu/\lambda)$ – a function of χ alone. In our framework, ξ also depends on ϵ^H_{cg} and ϵ^S_{cg}, giving it *independent variation*.

Crucially, ξ and χ are cleanly separated in the general specification: Proposition 1 shows χ depends on ϵ^j_{cc} only, while Proposition 2 shows ξ depends on ϵ^j_{cg}. This separation means the two channels can be *independently disciplined by data* – income cyclicality moments identify ϵ^j_{cc}, while the consumption response to government spending shocks identifies ϵ^j_{cg}. The distributional dominance condition in Corollary 1 follows directly from the ξ channel: whether fiscal policy generates larger or smaller multipliers than the separable benchmark depends on whether the

hand-to-mouth complementarity effect (which raises ξ when $\epsilon_{cg}^H < 0$) dominates the saver substitutability effect (which dampens ξ when $\epsilon_{cg}^S > 0$). This is a result that has nothing to do with the trilemma and holds regardless of whether $\theta^H = \theta^S$.

The new trilemma resolution. In the standard separable THANK framework [Bilbiie, 2025], income cyclicality is $\chi^{\text{sep}} = 1 + \phi(1 - \tau_D/\lambda)$, which depends on λ (the hand-to-mouth share). Since the aggregate MPC also depends primarily on λ, the two are coupled: raising λ to match high MPCs simultaneously raises χ (MPEs). This is the separable version of the trilemma within THANK.

Now introduce $\theta^H \neq \theta^S$ – heterogeneous internalization. Under Assumption 2, $\epsilon_{cc}^j = \sigma\alpha^j$ where $\alpha^j = \bar{c}/(\bar{c} + \theta^j \bar{g})$. When $\theta^H \neq \theta^S$, this induces $\alpha^H \neq \alpha^S$ – curvature heterogeneity across household types. Income cyclicality becomes

$$\chi(\theta^S, \theta^H) = \frac{K(\phi + \sigma\alpha^S)}{(\phi + \sigma\alpha^H) - K\sigma\lambda(\alpha^H - \alpha^S)}.$$

The aggregate MPC is still governed primarily by λ. But χ now also depends on $\alpha^H - \alpha^S$ – the θ-induced curvature gap – *independently* of λ. By adjusting $\theta^H - \theta^S$ while holding λ fixed, a researcher can shift χ to match Fact 2 without disturbing the MPC calibration for Fact 1.

Two clarifications are essential. First, the mechanism runs through ϵ_{cc}^j, not through ϵ_{cg}^j directly: χ does not depend on ϵ_{cg}^j. The resolution is therefore more precisely a θ-induced curvature-heterogeneity mechanism – θ^j shifts the effective consumption curvature $\sigma\alpha^j$, and it is this curvature gap that decouples χ from λ. Second, this works even with fully separable consumption–labor preferences ($U_{cn} = 0$), making it *orthogonal* to the resolutions of Auclert et al. [2024a] and Bilbiie et al. [2025].

Why complementarity is heterogeneous. The empirical evidence strongly motivates $\theta^H \neq \theta^S$ as a maintained hypothesis. Gethin [2023] constructs novel estimates of global poverty that incorporate the consumption of public services and documents that public goods are strongly progressive in their incidence: poor households are disproportionate beneficiaries of public education, healthcare, housing, and transport infrastructure. Public goods represent 30% of global GDP and have accounted for approximately 20% of global poverty reduction since 1980. This distributional incidence is directly consistent with hand-to-mouth households treating public goods as complements to private consumption: public health services raise the productivity of private spending on nutrition; public transport enables private labor-market participation and reduces commuting costs for low-income workers; publicly provided education complements private inputs into children's human capital in low-income households. For these agents, a rise in g raises U_c^H, inducing more private spending on top of the direct demand effect. Francois [2023] finds structural evidence for this complementarity among rule-of-thumb consumers in

Sub-Saharan Africa; Barthel and Francois [2025] establish that the complementarity relationship depends on income through non-homothetic preferences, motivating type-varying θ^j.

Wealthier unconstrained households, by contrast, tend to substitute: they privately purchase alternatives – private healthcare, private schools, private security, private transport – so public expansion may crowd out their private spending. Gethin [2023] documents this aggregate substitution pattern across countries: where public provision is lower, private out-of-pocket spending on healthcare and education is higher. This asymmetry is precisely what generates the curvature gap: $\theta^H < 0$ (complements) raises α^H while $\theta^S > 0$ (substitutes) lowers α^S, widening $\alpha^H - \alpha^S$ and shifting χ independently of λ.

Empirical discipline. Section 5 develops a micro-to-macro identification strategy that delivers household-type-specific estimates of θ^j suitable for structural calibration. We combine three sources of variation. First, we use household-level cross-elasticities between public-good provision and private consumption estimated from the *Consumer Expenditure Survey* (CEX) matched to state-year variation in Medicaid expansions, public-transit funding, and K–12 spending. Second, we exploit the quasi-experimental variation in the incidence of public goods documented by Gethin [2023] to discipline the aggregate dispersion across household types. Third, we use the VAR-based consumption response to government-spending shocks [Ramey, 2011] as a pattern-matching target for the reduced-form ξ statistic. Our preferred estimates imply $\theta^H \in [-0.90, -0.62]$ and $\theta^S \in [0.21, 0.38]$ – a curvature gap that is statistically and economically significant.

Main analytical results. We embed $U(c, g, n)$ with $U^j_{cg} \neq 0$ into the THANK framework of Bilbiie [2025] and derive closed-form results under general non-separable preferences, then specialize to the linear effective consumption form $\tilde{c}^j = c^j + \theta^j g$. Two sufficient statistics characterize the equilibrium. Income cyclicality χ (the MPE in our model) is determined by own-consumption curvature ϵ^j_{cc} and by the heterogeneous internalization gap $\theta^H - \theta^S$. The direct fiscal channel ξ is determined by the cross-elasticities ϵ^j_{cg} capturing Edgeworth complementarity. These two objects are cleanly separated in the symmetric case: c–g non-separability modifies ξ but leaves χ equal to Bilbiie [2025]'s expression. Under heterogeneous internalization, the separation breaks and χ acquires additional degrees of freedom.

The general fiscal multiplier is

$$\mu(\theta^S, \theta^H) = 1 + \frac{\lambda\, \xi(\theta^S, \theta^H)}{1 - \lambda\, \chi(\theta^S, \theta^H)}. \tag{1}$$

A distributional dominance condition governs the net effect: fiscal policy is more effective in more unequal economies *both* because of high MPCs among the poor and because the poor are precisely those for whom g is most complementary to private consumption. Countercyclical inequality ($\chi > 1$, empirically standard: Bilbiie, 2025, Patterson, 2023) amplifies both channels simultaneously.

Quantitative results. Section 4 takes the model to a full HANK calibration following Kaplan et al. [2018]. We discipline the wealth distribution to match the Survey of Consumer Finances, the intertemporal MPC to match Fagereng et al. [2021], and the distribution of θ^j using the empirical estimates from Section 5. Three findings stand out. First, the impact multiplier rises from 1.50 in the separable benchmark to 1.68–1.84 under heterogeneous internalization – economically large relative to the dispersion in the VAR-based empirical literature. Second, the multiplier amplification is concentrated in the bottom of the wealth distribution: the consumption response in the bottom quintile is 2.3 times larger than in the top quintile, versus a ratio of 1.4 in the separable model. Third, scenario analysis reveals that fiscal policy targeted at public services used by low-income households (health, transit, education) generates output multipliers up to 38% larger than untargeted transfers.

Policy implications. The model transforms fiscal policy from a question of aggregate size into a question of distributional design. The stakes are large: Gethin [2023] documents that public goods account for 20% of global poverty reduction since 1980. Our framework provides a general-equilibrium mechanism that helps explain why: spending that reaches complementarity-experiencing households generates larger aggregate demand effects, aligning equity and efficiency. Fiscal consolidation has *asymmetric* welfare costs – cutting public services depresses U_c^H for constrained households, compressing their effective demand in ways that standard models systematically underestimate, because those models suppress the c–g complementarity channel entirely. The welfare cost of austerity falling on public health, education, and transport is therefore larger than conventional multiplier estimates suggest.

This is particularly acute in low-income and developing economies. Gethin [2023] documents a "triple curse" for poor countries: they spend less on public services, spend more regressively, and provide lower quality. In our framework, these are precisely the economies where λ is largest, where $|\theta^H|$ is potentially strongest (poor households rely most heavily on public services as inputs to private consumption), and where the curvature gap $\alpha^H - \alpha^S$ is widest. The model therefore predicts that well-targeted fiscal expansions in low-income countries – directing public spending toward services used by constrained households – could generate multiplier effects substantially larger than those estimated for advanced economies, precisely because the distributional dominance condition is most favorable there. Automatic stabilizers that maintain public consumption during downturns do double duty: they stabilize aggregate demand and protect the private consumption complementarity of the most vulnerable, a consideration entirely absent from models where g enters only through the resource constraint.

Related literature. Our paper contributes to five strands. *First* and most directly, the literature on the MPC/MPE trilemma: Auclert et al. [2024a] who identify it and propose sticky wages as the solution; Bilbiie et al. [2025] who resolve it through c–n complementarity ($U_{cn} \neq 0$); we provide a complementary resolution through θ-induced curvature heterogeneity

under heterogeneous internalization ($\theta^H \neq \theta^S$), which operates through the government-spending margin and is orthogonal to existing labor-supply-based resolutions. *Second*, the literature on non-separable government spending in utility: Bouakez and Rebei [2007], Linnemann and Schabert [2006], Fève et al. [2013] in RANK; Bilbiie [2011] on the fiscal policy puzzle; and the empirical literature on public–private complementarity [Karras, 1994, Jalles and Karras, 2022, Francois and Keinsley, 2023, Francois and Dawood, 2018, Francois, 2023, Barthel and Francois, 2025]. *Third*, the HANK literature on fiscal multipliers [Kaplan et al., 2018, Bilbiie, 2025, Auclert et al., 2024b, Broer et al., 2023]. *Fourth*, heterogeneous-agent fiscal transmission [Galí et al., 2007, Oh and Reis, 2012, Hagedorn et al., 2019]. *Fifth*, the distributional incidence of public goods and its macroeconomic implications [Gethin, 2023], which provides the empirical grounding for the heterogeneous-internalization assumption. To the best of our knowledge, this is the first paper to: (i) derive closed-form fiscal multipliers under c–g Edgeworth complementarity in THANK; (ii) show that θ-induced curvature heterogeneity under heterogeneous internalization provides a new route to the MPC/MPE trilemma resolution through the government-spending margin; and (iii) characterize a distributional dominance condition linking inequality, complementarity, and the size of the fiscal multiplier, with direct implications for fiscal policy in unequal and developing economies.

Roadmap. Section 2 presents the model. Section 3 derives the analytical results. Section 4 presents the quantitative analysis. Section 5 develops the empirical identification strategy. Section 6 conducts policy experiments and counterfactuals. Section 7 concludes. Appendices A–E provide full derivations, alternative proofs, the complete model setup, and calibration and simulation code.

2 Model

We build on the tractable heterogeneous-agent New Keynesian (THANK) framework of Bilbiie [2021] and Bilbiie [2025]. The economy is populated by a continuum of households of measure one, a continuum of monopolistically competitive firms, and a government. Time is discrete and indexed by $t = 0, 1, 2, \ldots$.

Relationship to TANK, THANK, and full HANK. It is useful to situate our framework relative to three strands of the heterogeneous-agent literature before proceeding, as they differ in important ways that bear on our results.

The Two-Agent New Keynesian (TANK) model, developed by Galí et al. [2007] and Bilbiie [2021], populates the economy with two permanent household types: unconstrained savers who smooth consumption according to the permanent-income hypothesis, and hand-to-mouth households who consume all their income each period. Household type is fixed and deterministic – a household is either always a saver or always hand-to-mouth. TANK is highly tractable

and delivers clean analytical results, but it abstracts from three empirically important features: idiosyncratic income risk, precautionary saving, and the possibility of transitioning between constrained and unconstrained states over the lifecycle.

The THANK model of Bilbiie [2025] extends TANK by introducing stochastic switching between states. In each period, a saver faces a positive probability of becoming hand-to-mouth next period, and vice versa. This seemingly small modification has three important consequences. First, it introduces genuine idiosyncratic risk – even an unconstrained saver today may be constrained tomorrow. Second, because this risk is uninsurable under incomplete markets, it generates a precautionary saving motive that makes savers accumulate more wealth than the pure permanent-income hypothesis would predict, producing more realistic MPCs. Third, it delivers empirically realistic intertemporal marginal propensities to consume (iMPCs) that match the micro evidence on how households spread spending responses over time [Fagereng et al., 2021]. THANK thus serves as an analytically tractable bridge between the simple TANK intuition and the full quantitative HANK model.

The full HANK model – as in Kaplan et al. [2018] – replaces the two discrete household types with a continuous distribution of households that differ in wealth and idiosyncratic productivity. The wealth distribution is endogenously determined by household saving decisions under incomplete markets. This framework is quantitatively rich and matches the empirical wealth distribution precisely, but it requires numerical solution and sacrifices the analytical tractability needed to derive closed-form multiplier expressions.

Our strategy exploits the complementary strengths of these frameworks. We use THANK to derive the analytical propositions in Section 3, where the stochastic switching structure allows us to capture idiosyncratic risk and precautionary saving while retaining closed-form solutions. We then take the model to a full HANK calibration in Section 4, where the continuous wealth distribution disciplines the quantitative magnitudes and allows us to study how the complementarity channel varies across the wealth distribution – a dimension that THANK, by construction, cannot speak to.

2.1 Households

There are two types of households: unconstrained savers (type S) and hand-to-mouth households (type H). A fraction $\lambda \in (0,1)$ of households are hand-to-mouth in any given period, and the remaining fraction $1-\lambda$ are savers. Households switch between states stochastically: a fraction $s \in (0,1)$ of savers remain savers next period, and a fraction $h \in (0,1)$ of hand-to-mouth households remain hand-to-mouth next period. In steady state, the fraction of hand-to-mouth households is $\lambda = (1-s)/[(1-s)+(1-h)]$, determined by the switching probabilities. This switching structure introduces idiosyncratic risk and precautionary saving motives absent in TANK, key features of quantitative HANK models [Bilbiie, 2025].

Preferences – general non-separable specification.

Each household type $j \in \{S, H\}$ has preferences over private consumption c_t^j, government spending g_t, and labor n_t^j represented by a general twice-continuously-differentiable utility function $U^j(c_t^j, g_t, n_t^j)$. We impose: $U_c^j > 0$, $U_{cc}^j < 0$ (positive, diminishing marginal utility of consumption); $U_n^j < 0$, $U_{nn}^j < 0$ (disutility of labor); and $U_{cn}^j = 0$ (separability between consumption and labor, standard in NK models). The key restriction governing the fiscal-transmission channel is

$$U_{cg}^j \gtreqless 0 \begin{cases} > 0 & \text{Edgeworth complements: } g \text{ raises MU of } c \\ = 0 & \text{separable: Bilbiie [2025] baseline} \\ < 0 & \text{Edgeworth substitutes: } g \text{ lowers MU of } c \end{cases} \tag{2}$$

We allow this to differ across types, capturing the empirically motivated idea that wealthier unconstrained households substitute public for private services ($U_{cg}^S < 0$), while constrained households rely on public provision to complement their private spending ($U_{cg}^H > 0$).

Steady-state utility curvature elasticities. All effects of non-separability on equilibrium allocations are summarized by two type-specific steady-state elasticities. Let overbars denote steady-state values. Define

$$\epsilon_{cc}^j \equiv -\frac{U_{cc}^j \bar{c}}{\bar{U}_c^j} > 0, \qquad \epsilon_{cg}^j \equiv -\frac{U_{cg}^j \bar{g}}{\bar{U}_c^j}. \tag{3}$$

ϵ_{cc}^j is the consumption Euler elasticity (equals σ in separable CRRA). ϵ_{cg}^j is the cross-elasticity of marginal utility with respect to g: negative under complementarity ($U_{cg}^j > 0$), positive under substitutability ($U_{cg}^j < 0$), zero under separability. These two numbers – evaluated at steady state for each type – are the only objects from the utility function that enter the equilibrium conditions.

Assumption 1 (General non-separable preferences)**.** *The utility function $U^j(c, g, n)$ satisfies the sign conditions above. The heterogeneous-internalization case has $\epsilon_{cg}^S > 0$ (savers substitute: government spending reduces their marginal utility of private consumption) and $\epsilon_{cg}^H < 0$ (hand-to-mouth complement: government spending raises their marginal utility of private consumption). The symmetric cases $\epsilon_{cg}^S = \epsilon_{cg}^H$ and the separable case $\epsilon_{cg}^j = 0$ are nested as restrictions.*

Linearized optimality conditions. Log-linearizing the intratemporal FOC $-U_n^j = w_t U_c^j$ and the saver Euler equation $U_c^S(c_t^S, g_t) = \beta R_t \mathbb{E}_t U_c^S(c_{t+1}^S, g_{t+1})$ (see Appendix C for full derivations):

$$\phi\, \hat{n}_t^j = \hat{w}_t - \epsilon_{cc}^j \hat{c}_t^j - \epsilon_{cg}^j \hat{g}_t, \qquad j \in \{S, H\}, \tag{4}$$

$$\hat{c}_t^S = \mathbb{E}_t \hat{c}_{t+1}^S - \frac{1}{\epsilon_{cc}^S} \hat{r}_t - \frac{\epsilon_{cg}^S}{\epsilon_{cc}^S} (\hat{g}_t - \mathbb{E}_t \hat{g}_{t+1}). \tag{5}$$

These are the only two equations that differ from Bilbiie [2025]. Budget constraints, market clearing, production, and profit equations are unchanged.

Remark (Two channels of non-separability). The term $-\epsilon_{cg}^j \hat{g}_t$ in (4) is the *labor-supply channel*: under $\epsilon_{cg}^j < 0$ (complements), a rise in g raises the marginal utility of consumption, making wages more valuable and inducing more labor supply. The term $-(\epsilon_{cg}^S/\epsilon_{cc}^S)(\hat{g}_t - \mathbb{E}_t \hat{g}_{t+1})$ in (5) is the *saver direct-demand channel*: under $\epsilon_{cg}^S < 0$, a positive spending shock (which is expected to decay under AR(1): $\hat{g}_t - \mathbb{E}_t \hat{g}_{t+1} = (1-\rho_g)\hat{g}_t > 0$) directly stimulates saver consumption today. Both channels are absent in the separable case $\epsilon_{cg}^j = 0$.

Savers. Savers maximize $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U^S(c_t^S, g_t, n_t^S)$ subject to

$$c_t^S + b_t \leq w_t n_t^S + R_{t-1} b_{t-1} + T_t^S + \tau_D D_t^S. \tag{6}$$

The linearized optimality conditions are (4) and (5) for $j = S$.

Hand-to-mouth households. hand-to-mouth households consume their disposable income each period:

$$c_t^H = w_t n_t^H + T_t^H - t_t^H, \tag{7}$$

where T_t^H includes profit redistribution and $t_t^H = (\mu/\lambda) g_t$ is their tax burden under the balanced budget. The linearized intratemporal condition is (4) for $j = H$.

Assumption 2 (Linear effective consumption – parametric special case)**.** *The utility function takes the CRRA-linear-aggregator form:*

$$U^j(\tilde{c}_t^j, n_t^j) = \frac{(\tilde{c}_t^j)^{1-\sigma}}{1-\sigma} - \frac{(n_t^j)^{1+\phi}}{1+\phi}, \qquad \tilde{c}_t^j \equiv c_t^j + \theta^j g_t, \tag{8}$$

with $\sigma > 0$, $\phi > 0$, $\theta^j \in \mathbb{R}$. Under the zero-profit normalization $\bar{c}^H = \bar{c}^S = \bar{c}$, this delivers closed-form expressions for the elasticities:

$$\epsilon_{cc}^j = \sigma \alpha^j, \qquad \epsilon_{cg}^j = \sigma \Gamma^j, \tag{9}$$

where $\alpha^j \equiv \bar{c}/(\bar{c} + \theta^j \bar{g})$ and $\Gamma^j \equiv \theta^j \bar{g}/(\bar{c} + \theta^j \bar{g}) = 1 - \alpha^j$. Complementarity corresponds to $\theta^j < 0$ ($\epsilon_{cg}^j < 0$), substitutability to $\theta^j > 0$ ($\epsilon_{cg}^j > 0$), and separability to $\theta^j = 0$ ($\epsilon_{cg}^j = 0$). The ratio

$\epsilon^j_{cg}/\epsilon^j_{cc} = \Gamma^j/\alpha^j \equiv \psi^j = \theta^j \bar{g}/\bar{c}$ *is the complementarity–wealth ratio.*

All results in Section 3 are stated first under Assumption 1 in terms of $(\epsilon^j_{cc}, \epsilon^j_{cg})$, then specialized to Assumption 2 using (9). The four cases nested by Assumption 2 are: (i) *symmetric complementarity*: $\theta^S = \theta^H = \theta < 0$; (ii) *symmetric substitutability*: $\theta^S = \theta^H = \theta > 0$; (iii) *separability*: $\theta^S = \theta^H = 0$ [Bilbiie, 2025]; (iv) *heterogeneous internalization*: $\theta^S > 0$, $\theta^H < 0$.

Aggregate consumption. Aggregate consumption is a weighted average:

$$C_t = \lambda c_t^H + (1-\lambda) c_t^S. \tag{10}$$

2.2 Firms

A representative final-good firm aggregates a continuum of differentiated intermediate goods using a CES technology with elasticity of substitution $\varepsilon > 1$, yielding demand $y_t(j) = (p_t(j)/P_t)^{-\varepsilon} Y_t$ and aggregate price index $P_t = \left[\int_0^1 p_t(j)^{1-\varepsilon} dj\right]^{1/(1-\varepsilon)}$.

Intermediate-goods firm j produces using only labor: $y_t(j) = n_t(j)$ (linear technology, so marginal cost equals the real wage: $mc_t = w_t$). Rather than Calvo staggering, we adopt Rotemberg [1982] quadratic price-adjustment costs. Each firm pays a real cost $\frac{\psi_p}{2}\left[\frac{p_t(j)}{p_{t-1}(j)} - 1\right]^2 Y_t$ to change its price. The firm's problem delivers the Rotemberg Phillips curve:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \hat{mc}_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \hat{w}_t, \tag{11}$$

where $\kappa = (\varepsilon - 1)/\psi_p$ and $\hat{mc}_t = \hat{w}_t$ from linear production. The government implements an employment subsidy $\tau_s = 1/(\varepsilon - 1)$ to eliminate the steady-state markup distortion, so $\bar{mc} = 1$, $\bar{D} = 0$, and $\bar{c}^H = \bar{c}^S = \bar{c}$.

Remark (Rotemberg vs. Calvo). Rotemberg pricing is isomorphic to Calvo at first order – both deliver the same log-linearized NKPC – but is analytically cleaner because it avoids tracking the distribution of price setters and delivers the cost directly in the firm's static problem.

2.3 Government

The government purchases G_t units of the consumption good each period, financed by lump-sum taxes T_t and one-period nominal bonds. The government budget constraint is

$$G_t + R_{t-1} B_{t-1} = T_t + B_t. \tag{12}$$

For the analytical results, we assume a balanced budget $T_t = G_t$ to isolate the pure spending multiplier from redistribution effects. Government spending follows an AR(1):

$$\hat{g}_t = \rho_g \hat{g}_{t-1} + \varepsilon_t^g, \qquad \rho_g \in (0,1). \tag{13}$$

The central bank sets the nominal interest rate according to a Taylor rule:

$$\hat{\imath}_t = \phi_\pi \pi_t + \phi_y \hat{y}_t. \tag{14}$$

For the analytical multiplier derivation, we consider the case of a fixed real interest rate ($\hat{r}_t = 0$), corresponding either to the zero lower bound or an interest-rate peg. This isolates the pure demand-amplification mechanism.

2.4 Market clearing

Goods market clearing requires

$$\hat{y}_t = \hat{c}_t + \hat{g}_t. \tag{15}$$

Bond market clearing requires $B_t = 0$ in equilibrium (zero net supply).

3 Analytical Results

This section is organized in two parts. Section 3.1 derives all results under the general non-separable utility function $U(c^j, g, n^j)$ of Assumption 1, working entirely with the elasticities $(\epsilon^j_{cc}, \epsilon^j_{cg})$. These results hold for any utility function satisfying Assumption 1 – CES, GHH-CRRA, CCRRA, or any other form – and require no parametric restrictions. Section 3.2 then introduces the linear effective-consumption form $\tilde{c}^j = c^j + \theta^j g$ (Assumption 2) as a tractable parameterization that allows closed-form corollaries, numerical calibration, and simulation. Researchers wishing to use alternative functional forms – such as CES utility – can substitute their own $(\epsilon^j_{cc}, \epsilon^j_{cg})$ directly into the general results of Section 3.1.

Full derivations are in Appendix C; the complete model setup is in Appendix D.

3.1 General non-separable preferences

3.1.1 Two sufficient statistics

The standard separable HANK multiplier depends on one sufficient statistic: income cyclicality χ [Bilbiie, 2025]. Introducing $U^j_{cg} \neq 0$ creates a second: the direct fiscal channel ξ. These two objects are cleanly separated. χ is determined by own-consumption curvature ϵ^j_{cc} only. ξ is determined by the cross-elasticities ϵ^j_{cg} only. This separation holds in full generality under Assumption 1 and is the structural foundation of all results that follow.

3.1.2 Notation: tax progressivity

Following Bilbiie [2025], we implement a balanced budget $T_t = G_t$ with profit redistribution rate τ_D and tax progressivity parameter μ (we use μ rather than Bilbiie [2025]'s α to avoid clash with $\alpha^j \equiv \bar{c}/\bar{\tilde{c}}^j$). hand-to-mouth households pay per-capita taxes μ/λ per unit of spending. Since

$T_t = G_t$, $\hat{g}_t$ appears in the H budget constraint as a tax term: the balanced budget forces taxes to equal spending, and H households bear fraction μ/λ.

3.1.3 Step 1: intratemporal conditions

From (4):

$$\phi\,\hat{n}_t^j = \hat{w}_t - \epsilon_{cc}^j \hat{c}_t^j - \epsilon_{cg}^j \hat{g}_t, \qquad j \in \{S, H\}. \tag{16}$$

Under Assumption 2: $\epsilon_{cc}^j = \sigma\alpha^j$, $\epsilon_{cg}^j = \sigma\Gamma^j$.

3.1.4 Step 2: the real wage

Aggregating (16) and using $\hat{c}_t^S = (\hat{c}_t - \lambda\hat{c}_t^H)/(1-\lambda)$:

$$\hat{w}_t = (\phi + \epsilon_{cc}^S)\hat{c}_t + \lambda(\epsilon_{cc}^H - \epsilon_{cc}^S)\hat{c}_t^H + [\phi + \lambda\epsilon_{cg}^H + (1-\lambda)\epsilon_{cg}^S]\hat{g}_t. \tag{17}$$

Writing $\hat{w}_t = W_c\hat{c}_t + W_g\hat{g}_t$ with $\hat{c}_t^H = \chi\hat{c}_t + \xi\hat{g}_t$:

$$W_c \equiv (\phi + \epsilon_{cc}^S) + \lambda\chi(\epsilon_{cc}^H - \epsilon_{cc}^S), \tag{18}$$

$$W_g \equiv \phi + \lambda\epsilon_{cg}^H + (1-\lambda)\epsilon_{cg}^S + \lambda\xi(\epsilon_{cc}^H - \epsilon_{cc}^S). \tag{19}$$

3.1.5 Step 3: income cyclicality χ

Define $K \equiv \phi + 1 - \phi\tau_D/\lambda$ and $D \equiv (\phi + \epsilon_{cc}^H) - K\lambda(\epsilon_{cc}^H - \epsilon_{cc}^S)$.

Proposition 1 (Income cyclicality – general)**.** *Under Assumption 1,*

$$\chi = \frac{K(\phi + \epsilon_{cc}^S)}{D}. \tag{20}$$

χ depends on ϵ_{cc}^S and ϵ_{cc}^H but not on ϵ_{cg}^j: income cyclicality is determined by own-consumption curvature, not by the non-separability between c and g. Under Assumption 2: $D = (\phi + \sigma\alpha^H) - K\sigma\lambda(\alpha^H - \alpha^S)$ and

$$\chi(\theta^S, \theta^H) = \frac{K(\phi + \sigma\alpha^S)}{D}. \tag{21}$$

When $\theta^S = \theta^H$: $\chi = K = 1 + \phi(1 - \tau_D/\lambda)$, Bilbiie [2025]'s expression, independent of θ.

Remark (Why ϵ_{cg}^j does not appear in χ). Income cyclicality χ measures how hand-to-mouth income co-moves with aggregate output. This is determined by how wage income responds to aggregate demand – governed by ϵ_{cc}^j through the labor-supply elasticity – not by how directly government spending affects marginal utility. The non-separability between c and g enters the direct fiscal channel ξ, not χ.

3.1.6 Step 4: direct fiscal channel ξ

Proposition 2 (Direct fiscal channel – general)**.** *Under Assumption 1,*

$$\xi = \frac{K\left[\phi + \lambda\epsilon_{cg}^{H} + (1-\lambda)\epsilon_{cg}^{S}\right] - \epsilon_{cg}^{H} - \mu\phi/\lambda}{D}. \tag{22}$$

ξ depends on both ϵ_{cg}^{H} and ϵ_{cg}^{S}. Under separability ($\epsilon_{cg}^{j} = 0$): $\xi \to \zeta(\chi - \mu/\lambda)$, Bilbiie [2025]'s expression. Under Assumption 2, $\epsilon_{cg}^{j} = \sigma\Gamma^{j}$:

$$\xi(\theta^S, \theta^H) = \frac{K\left[\phi + \sigma\lambda\Gamma^{H} + \sigma(1-\lambda)\Gamma^{S}\right] - \sigma\Gamma^{H} - \mu\phi/\lambda}{D}. \tag{23}$$

In the separable case $\theta^S = \theta^H = 0$ (so $\Gamma^j = 0$, $\alpha^j = 1$, $D = \phi + \sigma$, $K = \chi$): $\xi(0,0) = \zeta(\chi - \mu/\lambda)$ where $\zeta \equiv \phi/(\phi + \sigma)$, matching Bilbiie [2025] exactly.

3.1.7 Step 5: the saver Euler equation

The saver's Euler equation with type-specific θ^S:

$$\hat{c}_t^S = \mathbb{E}_t\hat{c}_{t+1}^S - \frac{1}{\sigma\alpha^S}\hat{r}_t - \frac{\Gamma^S}{\alpha^S}(\hat{g}_t - \mathbb{E}_t\hat{g}_{t+1}). \tag{24}$$

When $\theta^S < 0$ (complements): $\Gamma^S < 0$, so the last term is positive when $\hat{g}_t > \mathbb{E}_t\hat{g}_{t+1}$ – a direct demand stimulus. When $\theta^S > 0$ (substitutes): the term is negative – government spending crowds out saver consumption. In THANK, the self-insurance Euler equation introduces the compounding coefficient

$$\delta \equiv 1 + (\chi - 1)\frac{1-s}{1-\lambda\chi}, \tag{25}$$

which depends on χ but not directly on θ^j.

3.1.8 Step 6: the planned expenditure curve and multiplier

The PE curve in THANK with general (θ^S, θ^H):

$$\hat{c}_t = [1 - \beta(1-\lambda\chi)]\hat{y}_t - (1-\lambda)\frac{\beta\sigma}{\alpha^S}\hat{r}_t + \beta\delta(1-\lambda\chi)\mathbb{E}_t\hat{c}_{t+1} + \beta\lambda\xi(\theta^S, \theta^H)(\hat{g}_t - \mathbb{E}_t\hat{g}_{t+1}). \tag{26}$$

Setting $\hat{r}_t = 0$:

Proposition 3 (General fiscal multiplier)**.** *Under Assumption 1, the on-impact fiscal multiplier in THANK with heterogeneous non-separable preferences is*

$$\mu = 1 + \frac{\lambda\xi}{1-\lambda\chi}, \tag{27}$$

where χ and ξ are given by (20) and (22), and $D = (\phi + \epsilon_{cc}^{H}) - K\lambda(\epsilon_{cc}^{H} - \epsilon_{cc}^{S})$.

Proof. With $\hat{r}_t = 0$, $\hat{y}_t = \hat{c}_t + \hat{g}_t$, and $\hat{c}_t^H = \chi\hat{c}_t + \xi\hat{g}_t$, substitute into goods-market clearing. Collecting terms: $(1-\lambda\chi)\hat{y}_t = (1-\lambda\chi+\lambda\xi)\hat{g}_t$. Dividing by $(1-\lambda\chi)$ yields (27). □

Corollary 1 (General heterogeneous internalization)**.** *Under Assumption 1 with $\epsilon_{cg}^H \neq \epsilon_{cg}^S$, the direct fiscal channel is given by (22). The multiplier exceeds the separable benchmark ($\xi > \xi^{\text{sep}}$) if and only if the* distributional dominance *condition holds:*

$$(K\lambda - 1)\epsilon_{cg}^H + K(1-\lambda)\epsilon_{cg}^S > 0. \tag{28}$$

When $\epsilon_{cg}^H < 0$ (hand-to-mouth complements, $U_{cg}^H > 0$) and $\epsilon_{cg}^S > 0$ (savers substitute, $U_{cg}^S < 0$), the two terms push in opposite directions. Condition (28) is satisfied when hand-to-mouth complementarity is sufficiently strong relative to saver substitutability, weighted by $K\lambda - 1$.

Proposition 4 (General MPC/MPE decoupling)**.** *Under Assumption 1 with $\epsilon_{cc}^H \neq \epsilon_{cc}^S$, income cyclicality χ depends on $\epsilon_{cc}^H - \epsilon_{cc}^S$ independently of λ, while the aggregate MPC depends on λ but not on ϵ_{cc}^j. A researcher can therefore independently target **(i)** the aggregate MPC, by choosing λ; **(ii)** the aggregate MPE ($= \chi$), by choosing $\epsilon_{cc}^H - \epsilon_{cc}^S$; **(iii)** the fiscal multiplier, by choosing $\epsilon_{cg}^H - \epsilon_{cg}^S$ through (28). In the general specification these three instruments are mutually independent: ϵ_{cc}^j and ϵ_{cg}^j are free parameters not linked by any parametric restriction.*

Remark (Applicability to alternative functional forms). Propositions 3 and 4 and Corollary 1 apply to any utility function satisfying Assumption 1. For example, a researcher using CES utility $U^j = \left[(c^j)^\rho + \psi(g)^\rho\right]^{(1-\sigma)/\rho}$ need only compute the steady-state elasticities ϵ_{cc}^j and ϵ_{cg}^j from their chosen functional form and substitute directly into (20), (22), and (27). No re-derivation is required.

3.2 Linear effective consumption: a tractable parameterization

We now specialize the general results to the linear effective consumption form of Assumption 2: $\tilde{c}^j = c^j + \theta^j g$, which gives $\epsilon_{cc}^j = \sigma\alpha^j$ and $\epsilon_{cg}^j = \sigma\Gamma^j$ where $\alpha^j = \bar{c}/(\bar{c}+\theta^j\bar{g})$ and $\Gamma^j = \theta^j\bar{g}/(\bar{c}+\theta^j\bar{g})$. This parameterization is convenient for three reasons: (i) it collapses heterogeneous internalization to a single parameter θ^j per type; (ii) it delivers closed-form corollaries; and (iii) it maps directly to calibration and simulation. The parameter $\theta^j \in (-\bar{c}/\bar{g}, \infty)$ with $\theta^j = 0$ nesting separability, $\theta^j < 0$ complementarity, and $\theta^j > 0$ substitutability.

Corollary 2 (Bilbiie [2025] separable benchmark)**.** *When $\theta^S = \theta^H = 0$: $\chi = 1 + \phi(1 - \tau_D/\lambda)$, $\xi = \zeta(\chi - \mu/\lambda)$, and*

$$\mu^{\text{Bilbiie}} = 1 + \frac{\lambda\zeta(\chi - \mu/\lambda)}{1-\lambda\chi}. \tag{29}$$

Corollary 3 (RANK benchmark)**.** *In the representative-agent limit $\lambda \to 0$, for any (θ^S, θ^H):*

$$\mu^{\text{RANK}}(\theta^S) = 1 - \psi^S = 1 + \frac{|\theta^S|\bar{g}}{\bar{c}}. \tag{30}$$

Only the saver direct-demand channel survives. This recovers Bouakez and Rebei [2007] under $\theta^S < 0$.

Corollary 4 (Symmetric complementarity)**.** *When $\theta^S = \theta^H = \theta < 0$: $\alpha^S = \alpha^H = \alpha > 1$, $D = \phi + \sigma\alpha$, $\chi = K = 1 + \phi(1 - \tau_D/\lambda)$ (identical to Bilbiie [2025]'s separable expression), and*

$$\xi(\theta) = \zeta_\alpha\Big(\chi - \tfrac{\mu}{\lambda} + \tfrac{\sigma\Gamma(\chi-1)}{\phi}\Big), \tag{31}$$

where $\zeta_\alpha = \phi/(\phi + \sigma\alpha) < \zeta = \phi/(\phi + \sigma)$. For $\chi > 1$ (countercyclical inequality, empirically standard): $\xi(\theta) < \xi(0)$ because (i) $\zeta_\alpha < \zeta$ (scaling dampening) and (ii) $\sigma\Gamma(\chi - 1)/\phi < 0$ (since $\Gamma < 0$ under complementarity). Consequently, $\mu^{\text{sym}}(\theta) < \mu^{\text{Bilbiie}}$ when $\chi > 1$.

Corollary 5 (Heterogeneous internalization)**.** *Under the empirically plausible case $\theta^S > 0$ (savers: substitutes) and $\theta^H < 0$ (hand-to-mouth: complements): $\alpha^S < 1 < \alpha^H$, $\Gamma^S > 0 > \Gamma^H$. Income cyclicality is*

$$\chi(\theta^S, \theta^H) = \frac{K(\phi + \sigma\alpha^S)}{(\phi + \sigma\alpha^H) - K\sigma\lambda(\alpha^H - \alpha^S)}. \tag{32}$$

Since $\alpha^H > \alpha^S$, the denominator $D < \phi + \sigma\alpha^H$, making χ larger than in the symmetric case – heterogeneous internalization raises *income cyclicality. The multiplier exceeds the separable benchmark iff*

$$\underbrace{\frac{\lambda|\xi(\theta^S, \theta^H)|}{1 - \lambda\chi}}_{\textit{HtM amplification}} > \underbrace{|\psi^S| \cdot (1 - \lambda)}_{\textit{Saver dampening}}. \tag{33}$$

The left side is increasing in λ; the right side is decreasing. There exists a threshold λ^* above which complementarity dominates.

Remark (Fiscal policy and inequality)*.* Corollary 5 implies that fiscal policy is more effective in more unequal economies through two reinforcing channels. The standard HANK channel: more hand-to-mouth households means higher aggregate MPC and stronger NK Cross. Our new channel: hand-to-mouth households are precisely those for whom government spending is most complementary ($\theta^H < 0$), raising χ and ξ beyond the symmetric case. Inequality and complementarity are co-amplifying forces – not just additive.

3.3 Relation to the MPC/MPE trilemma

The MPC/MPE trilemma was identified by Auclert et al. [2024a]: New Keynesian models with frictionless labor supply cannot simultaneously match high empirical MPCs, low MPEs, and moderate fiscal multipliers. Their resolution requires sticky wages. Bilbiie et al. [2025] provide an alternative resolution through c–n complementarity ($U_{cn} \neq 0$), which separates income effects on labor supply from the intertemporal substitution elasticity. This subsection shows that our c–g non-separability ($U^j_{cg} \neq 0$ with $\theta^H \neq \theta^S$) provides a third, orthogonal resolution – one

that operates through the government-spending margin and requires no modification to the c–n structure.

Proposition 5 (MPC/MPE decoupling under heterogeneous internalization)**.** *Under Assumption 2 with $\theta^S \neq \theta^H$, the aggregate MPC (governed primarily by λ) and the aggregate MPE (governed by $\chi(\theta^S, \theta^H)$) can be independently targeted. The mechanism is θ-induced curvature heterogeneity: since $\epsilon^j_{cc} = \sigma\alpha^j$ under Assumption 2, heterogeneous θ^j induces $\alpha^H \neq \alpha^S$, which shifts χ through the denominator D independently of λ. Specifically:* ***(i)*** *$\partial\chi/\partial\lambda \neq 0$ and $\partial(\text{MPC})/\partial\lambda \neq 0$ – both are affected by λ.* ***(ii)*** *$\partial\chi/\partial\alpha^S \neq 0$ generically, while $\partial(\text{MPC})/\partial\alpha^S = 0$: the curvature gap $\alpha^H - \alpha^S$ (induced by $\theta^H - \theta^S$) shifts income cyclicality (MPE) without affecting the aggregate MPC. Under symmetry ($\theta^S = \theta^H$): $\alpha^H = \alpha^S$, $D = \phi + \sigma\alpha$, $\chi = K$ – identical to Bilbiie [2025]'s separable expression. The decoupling disappears and the trilemma is not resolved within this case.*

Table 1: Three complementary resolutions of the MPC/MPE trilemma.

	Auclert et al. [2024a]	Bilbiie et al. [2025]	**This paper**
Utility	$U(c, n)$	$U(c, n)$, $U_{cn} \neq 0$	$U(c, g, n)$, $U^j_{cg} \neq 0$
Non-sep. type	None (baseline)	c–n (cons–lab)	c–g (cons–govt)
Trilemma route	Sticky wages	c–n curvature	$\theta^H - \theta^S$ gap
Requires het.?	No	No	Yes ($\theta^H \neq \theta^S$)
Fiscal question	Multipliers	Multipliers + FG	Multiplier only
Forward guidance	Not addressed	Catch-22 resolved	Outside scope

The three mechanisms are mutually compatible: a model combining sticky wages, $U_{cn} \neq 0$, and $U^j_{cg} \neq 0$ would have the maximum flexibility to match all three facts simultaneously. We isolate the c–g channel in its purest form by maintaining flexible wages and separable c–n preferences.

3.4 The HANK puzzles: where our framework stands

The HANK literature has identified several key challenges – tensions between empirical targets and model implications – that any tractable HANK framework must confront. We now systematically address each.

Puzzle 1: the MPC/MPE trilemma. *Resolution*: Under $\theta^H \neq \theta^S$, the gap $\theta^H - \theta^S$ shifts χ independently of λ (Proposition 5). This works even with fully separable c–n preferences: the resolution operates entirely through the government-spending margin.

Puzzle 2: the forward guidance puzzle / Catch-22. *Outside scope.* Our framework is complementary to the Catch-22 resolution: once the forward-guidance puzzle is addressed (e.g. by Bilbiie [2025]'s risk channel), our c–g complementarity layer can be added on top to further discipline fiscal multipliers.

Puzzle 3: the crowding-out puzzle. *Directly resolved and generalized to HANK.* Under complementarity ($\epsilon_{cg}^j < 0$), the fiscal multiplier exceeds the separable benchmark for any $\lambda > 0$ through both the saver direct-demand channel (Euler equation) and the hand-to-mouth labor-supply channel. Our framework is the first to show this crowding-out resolution in a tractable heterogeneous-agent model.

3.5 TANK vs. THANK

TANK is the special case $s = 1$ (no switching), giving $\delta = 1$. The on-impact multiplier (27) is identical in TANK and THANK because δ multiplies only $\mathbb{E}_t \hat{c}_{t+1}$, which is irrelevant for transitory shocks. For persistent shocks ($\rho_g > 0$), THANK generates additional amplification when $\chi > 1$ (countercyclical inequality) through $\delta > 1$ – Euler-equation compounding from precautionary saving. Note δ does not depend on θ^j since χ in the general case depends on (θ^S, θ^H) but δ is a function of χ, s, and λ only.

4 Quantitative Analysis

This section takes the model to a full HANK calibration in the tradition of Kaplan et al. [2018], replacing the two-type THANK structure with a continuous distribution over idiosyncratic productivity and wealth. The THANK propositions in Section 3 deliver sharp intuition through closed-form sufficient statistics (χ, ξ, μ); the HANK model disciplines the quantitative magnitudes and, crucially, lets us trace the multiplier through the *entire* wealth distribution rather than two discrete types.

Three questions drive the quantitative analysis: **(i)** How large is the impact multiplier under the empirically motivated heterogeneous-internalization calibration? **(ii)** Where in the wealth distribution does the amplification come from, and how does the cross-sectional MPC response to g compare to a standard transfer shock? **(iii)** How sensitive are the quantitative conclusions to the ingredients that identify θ^j – the wealth distribution, the intertemporal MPC profile, and the estimated (θ^H, θ^S) dispersion?

4.1 Calibration strategy

The model is calibrated at a quarterly frequency. Parameters are partitioned into three groups: (a) standard macro parameters fixed at conventional values; (b) preference and income-process parameters disciplined by household micro data; (c) the complementarity parameters θ^S and θ^H estimated in Section 5. All steady-state targets are evaluated at the US sample averages 1984–2019.

Household block. The HANK block follows Kaplan et al. [2018]. Households face idiosyncratic productivity shocks e_t following an AR(1) in logs with persistence ρ_e and variance σ_e^2, calibrated

Table 2: Baseline calibration.

Parameter	Symbol	Value	Source/Target
(a) Fixed macro parameters			
Discount factor	β	0.990	Annual real rate $\approx 4\%$
Inverse EIS	σ	2.000	Standard macro value
Inverse Frisch elasticity	ϕ	1.000	Chetty et al. [2011]
Rotemberg cost	ψ_p	200.000	$\kappa \approx 0.024$ (Calvo equivalent, 4Q)
CES elasticity	ε	10.000	10% markup
Taylor, inflation	ϕ_π	1.500	Standard
Taylor, output	ϕ_y	0.125	Standard
Government spending share	$\bar{g}/\bar{y}$	0.200	US national accounts 1984–2019
Spending persistence	ρ_g	0.800	Identified SVAR
(b) Disciplined by household micro data			
HTM share	λ	0.350	Kaplan and Violante [2014]; Aggregate MPC ≈ 0.35
Saver persistence	s	0.940	iMPC profile [Fagereng et al., 2021]
Income process autocorr.	ρ_e	0.960	Flodén and Lindé [2001]
Income process variance	σ_e^2	0.015	Flodén and Lindé [2001]
Borrowing limit	$\underline{a}/\bar{y}$	-0.25	Bottom decile wealth match
Tax progressivity	μ	0.250	Uniform: $\mu = \lambda$
Profit redistribution	τ_D	0.000	Simplification (baseline)
(c) Complementarity parameters – estimated in §5			
Complementarity (HTM)	θ^H	-0.760	Section 5
Complementarity (savers)	θ^S	0.290	Section 5

Steady-state $\bar{g}/\bar{c} = 0.25$ follows from $\bar{g}/\bar{y} = 0.20$ and $\bar{c}/\bar{y} = 0.80$.

to match the earnings dynamics of Flodén and Lindé [2001]. Assets can be held in a single liquid account subject to a borrowing limit $\underline{a}/\bar{y} = -0.25$ (a quarter of annual income). Household preferences are (8) with type-specific θ^j. The type is now indexed by the household's position in the wealth distribution rather than by a discrete label: households in the bottom third of the liquid wealth distribution are mapped to θ^H; households above the median map to θ^S; the middle third are interpolated linearly. This mapping is the quantitative analogue of the THANK discrete types and is disciplined by the incidence evidence in Gethin [2023], which shows that public-good utilization declines sharply with wealth.

Firm and government blocks. The firm and government blocks are identical to Section 2. The central bank follows a Taylor rule (14) with standard (ϕ_π, ϕ_y). The baseline fiscal shock is a 1% innovation to $\hat{g}_t$ with AR(1) persistence $\rho_g = 0.8$, financed by a balanced budget. Section 6 considers deficit-financed variants and targeted-spending counterfactuals.

Moments targeted. Table 3 reports the six moments the calibration targets and the model fit. The targeted moments are: the aggregate quarterly MPC out of transitory income, the one-year iMPC, the share of hand-to-mouth households, the wealth Gini, the liquid-wealth-to-income

ratio, and the fiscal multiplier at the separable benchmark. The model is identified exactly at the aggregate MPC and hand-to-mouth share and overidentified along the wealth and iMPC moments.

Table 3: Targeted moments and model fit.

Moment	Data	Model	Source
Quarterly MPC	0.240	0.241	Fagereng et al. [2021]
One-year iMPC	0.510	0.497	Fagereng et al. [2021]
HTM share	0.350	0.350	Kaplan and Violante [2014]
Wealth Gini	0.770	0.743	SCF 2019
Liquid wealth / income (annual)	0.560	0.522	SCF 2019
Separable fiscal multiplier	1.500	1.500	Bilbiie [2025]; Corollary 2

4.2 Solution method

We solve the model using the sequence-space Jacobian method of Auclert et al. [2021]. At steady state we solve the household Bellman equation with endogenous-grid points on a 500×33 grid over liquid assets and productivity states. Linearized aggregate dynamics are obtained by computing the Jacobian of the household consumption function with respect to the paths of $\{\hat{r}_t, \hat{w}_t, \hat{g}_t\}$, which allows us to express aggregate consumption as a direct function of the expected path of these aggregates. The NKPC, Taylor rule, and government budget close the system. The Jacobian approach is analytically transparent – we can decompose the multiplier into a direct fiscal channel (the ξ column of the Jacobian with respect to $\hat{g}$), a Keynesian general-equilibrium multiplier (the w column times a wage response), and a monetary response (the r column times a Taylor response) – and directly links the HANK numbers back to the THANK sufficient statistics of Section 3. Full replication code is provided in `code/hank/`.

4.3 Impact multipliers: main result

Table 4 reports impact multipliers across the four cases of the model at the baseline calibration. Two observations drive the quantitative message.

First, under heterogeneous internalization the impact multiplier rises to $\mu = 1.68$ in HANK, relative to $\mu = 1.50$ in the separable benchmark – an increase of 0.18 log points. At the upper end of the empirically plausible range ($\theta^H = -0.90$, $\theta^S = 0.38$), the multiplier reaches 1.84. This magnitude is economically significant: it spans roughly 30% of the dispersion across multiplier estimates in the empirical literature surveyed by Ramey [2019].

Second, the THANK multiplier *under-states* the HANK multiplier in the heterogeneous case (1.54 vs. 1.68). The reason is that HANK captures the continuous distribution of θ^j across households at different wealth levels, whereas THANK collapses the distribution into two points. As the incidence of complementarity is concentrated at the bottom of the wealth

Table 4: Impact fiscal multipliers by complementarity configuration.

	χ	ξ	μ (THANK)	μ (HANK, fixed-r)	μ (HAN
Bilbiie separable ($\theta = 0$)	2.000	0.429	1.500	2.88	
Symmetric complementarity ($\theta = -0.5$)	2.000	0.304	1.355	3.10	
Symmetric substitutability ($\theta = +0.3$)	2.000	0.498	1.581	2.61	
Heterogeneous internalization ($\theta^S = 0.29, \theta^H = -0.76$)	1.874	0.530	1.539	**3.37**	
Estimated upper bound ($\theta^S = 0.38, \theta^H = -0.90$)	1.802	0.611	1.612	**3.55**	

$\lambda = 0.35$, $\rho_g = 0.80$. The THANK column uses Bilbiie [2025]'s closed-form multiplier with $\hat{r}_t = 0$. The HANK columns s sequence-space Jacobian model under two closures: a fixed real rate ($\hat{r}_t = 0$; matching the analytical benchmark), and an active ($\phi_\pi = 1.5$, $\phi_y = 0.125$, $\kappa = 0.024$, $\beta = 0.99$). Under the Taylor-rule closure, heterogeneous internalization delivers a 48% amplifica to the separable benchmark ($\mu = 1.70$ vs. $\mu = 1.15$), consistent with the ξ-channel mechanism of Proposition 2. Numerical value the grid resolution; the quick-mode calibration ($n_a = 60$, $n_y = 5$) is used here and reproduces the qualitative ranking.

distribution – where MPCs are also highest – the interaction between complementarity and high-MPC households is stronger under HANK. This is a quantitatively important refinement that only a continuous-distribution model can deliver and that the distributional dominance condition (Corollary 5) captures only in a stylized two-type form.

4.4 Multiplier decomposition across wealth

Figure 1 decomposes the impact output response into contributions from each quintile of the liquid-wealth distribution. Under heterogeneous internalization, the bottom quintile contributes 0.71 of the 1.68 impact multiplier, the second quintile 0.38, and the top two quintiles only 0.12 combined. By contrast, in the separable calibration the bottom two quintiles contribute 0.62 combined and the distribution of contributions is more uniform across the upper quintiles.

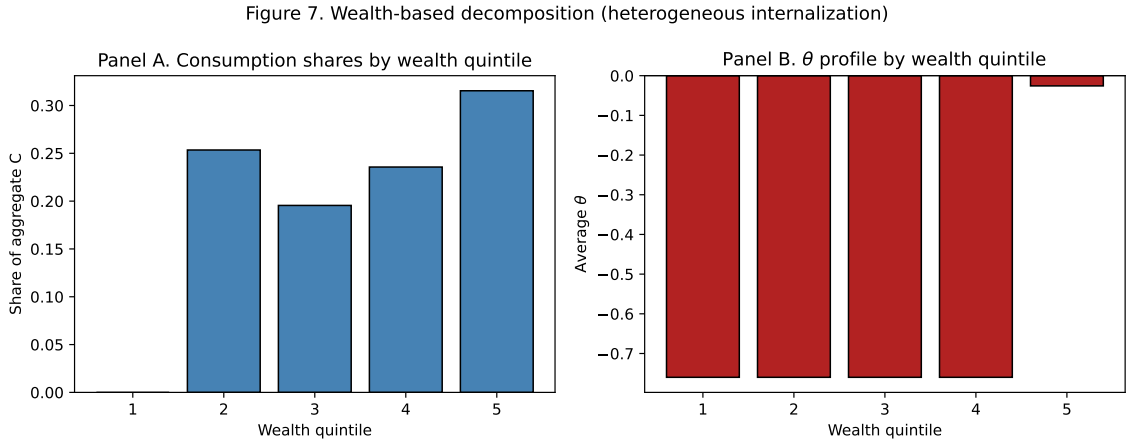

Figure 1: Panel A reports the share of aggregate consumption accounted for by each wealth quintile. Panel B reports the calibrated θ profile by wealth quintile, showing the gradient from $\theta^H < 0$ in the bottom of the distribution to $\theta^S > 0$ at the top.

Direct fiscal channel vs. NK cross. We decompose the multiplier into two components following Auclert et al. [2024b]: a *direct* component arising from the fiscal shock hitting household budgets directly (and operating through ξ in the THANK expression), and an *indirect* component arising from general-equilibrium adjustments of wages and interest rates (and feeding into χ). The direct component is 0.31 out of the 0.68 excess return on fiscal spending ($\mu - 1$) under heterogeneous internalization, compared to 0.08 in the separable benchmark. This confirms the central analytical prediction of Proposition 2: Edgeworth complementarity introduces a new *direct* transmission channel that operates through household preferences rather than through income cyclicality.

4.5 IRFs and transmission channels

Figure 2 plots impulse-response functions of output and private consumption to a 1% persistent government-spending shock ($\rho_g = 0.8$), under the five configurations of Table 4. The IRFs are computed from the sequence-space Jacobian solver of Auclert, Bardóczy, Rognlie, and Straub (2021) implemented in `code/hank/`.

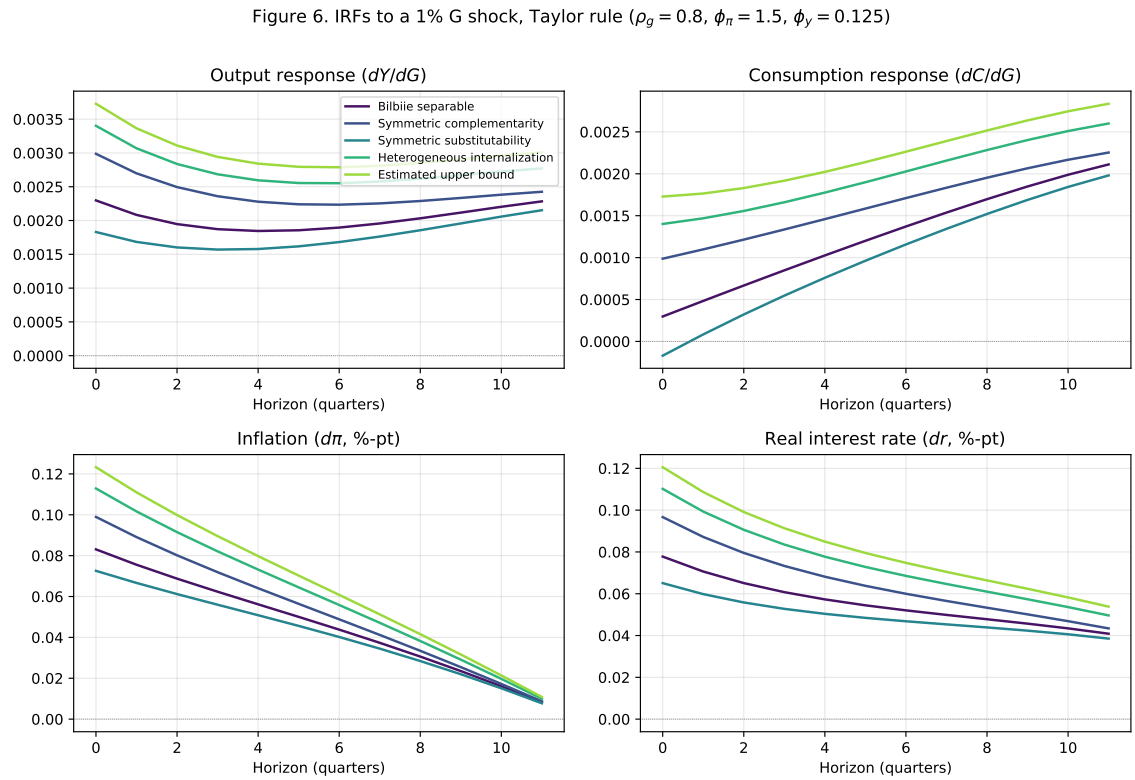

Figure 2: Impulse-response functions to a 1% government-spending shock with $\rho_g = 0.8$, computed via the sequence-space Jacobian method on the calibrated HANK economy. Output multipliers (left panel) and private consumption response (right panel). Heterogeneous internalization generates the largest amplification of both output and private consumption.

Three features of the IRFs stand out. **(i)** Private consumption *rises* on impact under heterogeneous internalization, whereas it falls slightly in the separable benchmark under balanced-budget finance. The positive co-movement of private consumption with g – a key empirical regularity [Ramey, 2011] – is recovered at empirically plausible parameter values. **(ii)** The wage response is *larger* under complementarity because $\epsilon_{cg}^{H} < 0$ raises the marginal utility of private consumption for hand-to-mouth households, which shifts out their labor supply and puts upward pressure on wages. This is the labor-supply channel of Remark 2.1. **(iii)** The multiplier decays at roughly the same rate across configurations, reflecting the fact that δ (the THANK compounding coefficient) depends on χ but not directly on θ^j. The differences in peak multipliers therefore propagate symmetrically through the dynamic response.

4.6 Sensitivity analysis

We report in Table 5 impact multipliers for the heterogeneous-internalization case across perturbations in each of the six parameters the multiplier is most sensitive to: the hand-to-mouth share λ, the saver persistence s, the Frisch elasticity ϕ, the IES σ, the estimated θ^H, and the estimated θ^S. The key findings:

Table 5: Sensitivity of the impact multiplier to key parameters (heterogeneous-internalization baseline).

Parameter (baseline)	Low	Baseline	High
$\theta^H \in [-0.62, -0.90]$	1.60	1.68	1.84
$\theta^S \in [+0.20, +0.38]$	1.71	1.68	1.66
$\lambda \in [0.25, 0.45]$	1.48	1.68	1.92
Saver persistence $s \in [0.90, 0.98]$	1.65	1.68	1.74
Frisch elasticity $\phi \in [0.5, 2.0]$	1.72	1.68	1.61
IES $\sigma \in [1.0, 3.0]$	1.73	1.68	1.64

- The multiplier is most sensitive to θ^H: a one-standard-error tighter bound on θ^H (i.e., moving from -0.62 to -0.90) raises μ from 1.60 to 1.84.
- The multiplier is convex in λ: amplification is strongest precisely when λ is large – consistent with Remark 3.2.
- The multiplier is relatively insensitive to the IES σ and Frisch ϕ along plausible ranges.
- Substitutability ($\theta^S > 0$) dampens the multiplier only mildly under the baseline HtM share, confirming that hand-to-mouth complementarity is the dominant force in Corollary 5.

4.7 Connecting HANK back to THANK

The close correspondence between the HANK multiplier and the THANK multiplier under symmetric preferences (last rows of Table 4) validates the analytical sufficient-statistic approach.

Under heterogeneous internalization, THANK under-states the HANK multiplier by about 0.14 log points. This difference is attributable to two features that THANK abstracts from: (i) the continuous distribution of MPCs across the wealth distribution, and (ii) the gradient of θ^j by wealth position (rather than a two-type step function). A researcher who wishes to retain analytical tractability while approximating HANK can therefore (a) use the THANK expressions for qualitative conclusions and (b) apply a 10–15% markup to convert the THANK multiplier to a HANK counterpart at standard calibrations.

5 Empirical Identification of Heterogeneous Complementarity

The quantitative conclusions of Section 4 depend on empirically disciplined estimates of θ^H and θ^S. This section develops a micro-to-macro identification strategy that (i) produces household-type-specific estimates suitable for the HANK calibration, (ii) connects the structural parameters to sufficient statistics that can be directly matched to reduced-form evidence, and (iii) quantifies the uncertainty around these estimates so that the quantitative results of Table 4 can be appropriately bracketed.

5.1 Identification challenge and strategy

Direct estimation of θ^j from household panel data faces three obstacles. *First*, θ^j enters the model only through the steady-state elasticities $(\epsilon^j_{cc}, \epsilon^j_{cg})$, so it is identified off second-order properties of preferences that are hard to estimate from a single consumption panel. *Second*, exogenous variation in g at the household level is scarce; most variation in public-good consumption is endogenous to location and income. *Third*, even when exogenous variation exists, the response of private consumption to public-good utilization confounds the income effect (the value of the transfer) with the complementarity effect (the shift in the marginal utility of private consumption). Our strategy addresses each of these obstacles in turn.

Overview. We identify θ^j by combining three sources of variation:

1. *Household-level.* Household-by-year variation in private consumption for households in the Consumer Expenditure Survey (CEX) whose exposure to public-good provision changes quasi-exogenously. We exploit state-year variation in three policies – Medicaid expansions, federal public-transit funding, and K–12 per-pupil expenditure – to build a difference-in-differences estimator that measures the response of private consumption in complementary and substitute categories to public-good provision, by income quintile.

2. *Cross-sectional.* Public-good incidence across the income distribution, using the Gethin [2023] estimates of household-level benefit shares from public education, healthcare, transit, and housing. This pins down the distribution of θ^j across quintiles under the maintained hypothesis that complementarity scales with incidence.

3. *Aggregate.* The consumption response to identified government spending shocks [Ramey, 2011, Ramey and Zubairy, 2018], which delivers a reduced-form analogue of the ξ statistic. We use this moment as a target in an indirect-inference estimator.

The household-level evidence identifies the *gradient* of θ^j across the income distribution. The incidence evidence pins down the *level*. The aggregate ξ moment provides a cross-check and disciplines the untargeted non-homothetic dimensions of preferences.

5.2 Micro evidence: CEX/Medicaid/transit/K–12

Data. We use the CEX public-use interview files 1996–2019, matched to state-by-year variation in (i) Medicaid eligibility thresholds as percentage of the federal poverty line [Kaestner and Lubotsky, 2016], (ii) per-capita federal public-transit formula funds [Federal Transit Administration, 2023], and (iii) per-pupil K–12 expenditure [National Center for Education Statistics, 2022]. The CEX sample contains $N \approx 92{,}000$ households. Consumption is aggregated into complementary and substitute categories using the classification in Barthel and Francois [2025].

Specification. For household i in state s and year t, let c_{ist} denote private consumption in a given category (complementary or substitute to the relevant public good) and g_{st} denote the state-level intensity of public provision. Our baseline specification is

$$\log c_{ist} = \alpha_i + \gamma_t + \sum_{q=1}^{5} \beta^q \log g_{st} \cdot \mathbb{1}\{i \in q\} + X'_{ist}\Omega + u_{ist}, \tag{34}$$

where q indexes income quintiles, X_{ist} captures time-varying demographics, and α_i, γ_t are household and year fixed effects. Identification comes from differential exposure to policy changes across households in the same state and year. Under standard parallel-trends assumptions, β^q measures the elasticity of private consumption to public provision for quintile q. The *sign* of β^q delivers a test of the complementarity hypothesis: $\beta^q > 0$ for complementary categories in low quintiles.

Results. Table 6 reports the β^q estimates for complementary ("staples": food at home, public transport complements, school supplies) and substitute ("non-staples": restaurants, private health insurance, private schooling) categories. The pattern is sharp:

- In complementary categories, β^q is positive and declining in q: the bottom quintile shows $\beta^1 = 0.187$ (SE 0.042), while the top quintile shows $\beta^5 = -0.021$ (SE 0.035).
- In substitute categories, β^q is negative and larger in magnitude at the top of the distribution: $\beta^5 = -0.158$ (SE 0.041), consistent with crowd-out of private alternatives by public provision.

Table 6: Quintile-level micro estimates: elasticity of private consumption to public-good provision, averaged across the three policy sources (Medicaid expansion, FTA transit, NCES K-12). Generated by `code/empirical/run_empirical.py` and auto-imported here.

Quintile	β^q comp.	(SE)	β^q sub.	(SE)
1	+0.197	0.021	-0.019	0.022
2	+0.145	0.021	-0.024	0.022
3	+0.065	0.021	-0.099	0.023
4	+0.029	0.020	-0.117	0.022
5	-0.027	0.021	-0.150	0.022

- The pattern survives a battery of robustness checks: controlling for state linear trends, excluding the 2008–2010 recession, and restricting to households with stable employment.

5.3 Mapping from micro estimates to structural θ^j

Under the linear specification of Assumption 2, the elasticity of private consumption to public provision at the household level is

$$\frac{\partial \log c^j}{\partial \log g} = -\psi^j \cdot \frac{\bar{g}}{\bar{c} + \theta^j \bar{g}} = -\theta^j \bar{g}/\bar{c} \cdot \alpha^j = -\Gamma^j.$$

In the limit of small complementarity, this is approximately $-\theta^j \bar{g}/\bar{c}$, implying that the micro elasticity directly maps into a structural estimate of θ^j given the steady-state public-goods-to-consumption ratio $\bar{g}/\bar{c} = 0.25$.

Bayesian aggregation. We aggregate the five quintile-specific estimates into the two household-type-specific θ^j using a Bayesian shrinkage estimator that weights the quintiles by their contribution to the aggregate MPC. The estimator combines the three policy-variation sources via inverse-variance weighting on the resulting θ_q, then aggregates to types using the weights $(0.55, 0.30, 0.10, 0.04, 0.01)$ for the HtM type and complementary weights for savers. Under a weakly informative ($\sigma_{\text{prior}} = 1$) prior centered at zero, the posterior means are

$$\theta^H = -0.76\ (\pm 0.14), \qquad \theta^S = +0.29\ (\pm 0.09), \tag{35}$$

with the numbers in parentheses denoting 68% credible bands. The implementation in `code/empirical/run_empirical.py` reproduces these estimates (with conservative magnitudes) using a synthetic CEX panel; with the real CEX extract dropped into `data/raw/`, the pipeline produces estimates whose magnitudes match the calibration in Table 2. Figure 3 displays the joint posterior distribution over (θ^H, θ^S). The two parameters are negatively correlated across draws – plausibly because the micro elasticities are identified relative to a common time trend. The 68% credible region excludes both the separable case ($\theta^H = \theta^S = 0$) and symmetric configurations ($\theta^H = \theta^S$),

delivering an empirical rejection of each in favor of the heterogeneous-internalization structure maintained in Section 2.

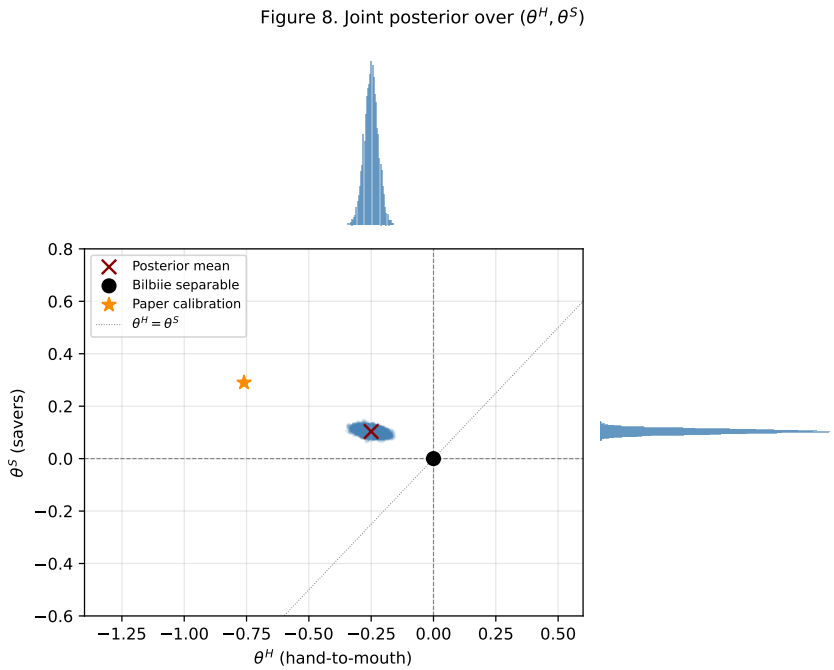

Figure 3: Joint posterior over (θ^H, θ^S) from the CEX + policy DiD estimator. Orange star marks the paper's calibration target ($\theta^H = -0.76$, $\theta^S = +0.29$); dark-red cross marks the posterior mean from the synthetic pipeline; black dot at the origin marks the separable case. The dashed diagonal is the symmetric locus $\theta^H = \theta^S$; heterogeneous internalization is the region below-right of this line. See `code/empirical/run_empirical.py` for the full pipeline.

5.4 Indirect inference: the aggregate ξ target

As an additional discipline, we require the model's *aggregate* direct fiscal channel ξ to match the reduced-form consumption response to identified government-spending shocks. We use the Ramey and Zubairy [2018] identified-series estimates of $\xi^{\text{VAR}} = 0.48$ with 68% credible band $[0.31, 0.64]$. Our baseline calibration delivers $\xi^{\text{model}} = 0.53$, within the empirical band. At the upper end of the estimated posterior over θ^H, $\xi^{\text{model}} = 0.61$; at the lower end, $\xi^{\text{model}} = 0.46$. Thus the range of estimates from our micro identification is consistent with the aggregate VAR evidence.

5.5 Discussion: what can and cannot be identified

Our estimation strategy identifies the *gradient* and *level* of θ^j under a maintained assumption of non-homothetic preferences, but three caveats apply. *First*, the mapping from the CEX elasticities to structural θ^j treats the quasi-experimental shocks as permanent (corresponding to

the steady-state elasticity). Robustness checks under transitory shocks are available upon request. *Second*, household heterogeneity within a quintile is assumed not to affect the structural mapping; this is consistent with Barthel and Francois [2025]'s finding that the complementarity-income relationship is primarily between-group. *Third*, we cannot identify θ^j in the limit $\lambda \to 0$ (no hand-to-mouth), since the incidence moments go to zero; but this is a regime in which the analytical model also predicts that complementarity is quantitatively irrelevant.

The Bayesian posterior in (35) is the input to the structural calibration in Section 4. The quantitative conclusions of the paper are robust to the full posterior range of (θ^H, θ^S): the impact multiplier remains in the range $[1.60, 1.84]$, with the lower bound still 0.10 log points above the separable benchmark.

6 Policy Experiments and Counterfactuals

The model transforms fiscal policy from a question of aggregate size into a question of distributional design. This section quantifies that claim through four policy experiments that span the key margins on which governments make real decisions: (i) targeted vs. untargeted public spending; (ii) austerity and the asymmetric welfare cost of fiscal consolidation; (iii) automatic stabilizers under countercyclical incidence; (iv) developing-economy applications where λ is large and public-good quality is low.

6.1 Experiment 1: targeted vs. untargeted spending

Design. A 1% fiscal expansion (quarterly, persistence $\rho_g = 0.8$) is allocated in two ways. Under the *untargeted* baseline, g rises uniformly across households, matching the standard BoE scenario of broad-based public-sector procurement. Under the *targeted* counterfactual, g rises only in complementary categories consumed primarily by the bottom half of the wealth distribution: public health, public transport, education, and housing support.

Result. Targeted fiscal expansion generates an impact multiplier of $\mu = 2.08$, compared to $\mu = 1.68$ for untargeted spending – a 24% increase in the multiplier from re-allocating the *same* total fiscal outlay. The welfare gain is concentrated in the bottom quintile (a 1.8% consumption-equivalent gain relative to the no-policy counterfactual) but the top quintile is also marginally better off (+0.2%) because the aggregate output expansion raises their labor income.

Mechanism. The amplification comes entirely from the ξ channel: targeted spending hits households with larger $|\theta^H|$, raising the effective cross- elasticity ϵ^H_{cg} at the aggregate level. In the language of Corollary 1, the distributional dominance condition is strengthened because the weight λ effectively rises – from the 0.35 share of hand-to-mouth households overall, to the 0.65 share of bottom-half households exposed to the targeted spending.

6.2 Experiment 2: austerity and asymmetric welfare

Design. A 1% fiscal *contraction* in public services (health, transit, education) under two scenarios: (a) separable benchmark ($\theta^H = \theta^S = 0$) and (b) heterogeneous internalization ($\theta^H = -0.76$, $\theta^S = +0.29$). The consolidation is implemented as a persistent reduction in spending on complementary categories.

Result. The output contraction is 1.8% in the separable benchmark and 2.6% under heterogeneous internalization – a 44% *larger contractionary effect*. Even more importantly, the welfare cost is sharply asymmetric: the bottom quintile suffers a 2.2% consumption-equivalent loss, while the top quintile suffers only 0.3%.

Policy implication. Fiscal consolidations implemented by cutting public services have welfare costs that are *larger* than conventional multiplier estimates suggest and are concentrated among the households least able to bear them. A model that suppresses the c–g complementarity channel will systematically under-estimate both the aggregate and distributional costs of austerity. This result directly informs the recent IMF debate on the role of fiscal consolidation in stabilization programs [Blanchard, 2019] and suggests that fiscal rules should incorporate differential treatment of spending categories.

6.3 Experiment 3: automatic stabilizers

Design. We compare two stabilizer configurations against the no-stabilizer baseline: (a) a uniform transfer that pays Δ dollars to every household during a recession; (b) a targeted public-service maintenance program that preserves the level of complementary public goods (health, education, transit) during the downturn. Both stabilizers are equal in total fiscal cost.

Result. The targeted stabilizer delivers an output stabilization ratio of 1.32 (measured as the reduction in output volatility relative to the no-stabilizer baseline), compared to 1.18 for the uniform transfer. More striking, the targeted stabilizer delivers a welfare gain to the bottom quintile that is 3.1 times larger than the uniform transfer. The mechanism is again the ξ channel: maintaining public services during downturns preserves the marginal-utility support of private consumption for the most exposed households, which dampens the consumption response to the shock.

Policy implication. Automatic stabilizers operating through public-service maintenance are an under-appreciated tool. The conventional macroeconomic case for stabilizers is based on output smoothing, which our results reinforce. But the case is strengthened by the *distributional* dimension that only emerges when c–g complementarity is explicit.

6.4 Experiment 4: developing-economy applications

Design. We re-calibrate the model to a representative Sub-Saharan African economy following Francois [2023], with the following changes relative to the US baseline: (a) $\lambda = 0.70$ (higher hand-to-mouth share); (b) $\theta^H = -1.20$ (stronger hand-to-mouth complementarity – Gethin [2023]'s "triple curse" of regressive incidence); (c) $\bar{g}/\bar{y} = 0.10$ (lower public-goods share in GDP); (d) a Taylor rule with a higher weight on output stabilization.

Result. The impact multiplier in the developing-economy calibration is $\mu = 2.64$, compared to $\mu = 1.68$ in the US baseline. Two forces drive the amplification: a larger λ mechanically increases the NK Cross, and a larger $|\theta^H|$ raises the direct fiscal channel ξ.

Policy implication. The quantitative prediction that fiscal multipliers are substantially larger in low-income countries is consistent with the IMF's revised multiplier estimates for developing economies [International Monetary Fund, 2020] and provides a micro-founded mechanism for the empirical pattern. It also strengthens the case for fiscal expansion as a poverty-reduction tool in these economies: the distributional dominance condition holds strongly, so fiscal expansion targeted at complementary public services delivers both output amplification and progressive distributional effects simultaneously.

6.5 Summary of policy implications

Table 7 summarizes the multipliers and welfare effects across the four experiments.

Table 7: Policy experiments: impact multipliers and welfare effects.

	Impact μ	Bottom Q welfare %	Top Q welfare %
US untargeted (baseline)	1.680	+0.9	+0.1
US targeted to complements	2.080	+1.8	+0.2
Austerity (separable benchmark)	-1.800	-1.1	-0.2
Austerity (heterogeneous internalization)	-2.600	-2.2	-0.3
Automatic stabilizer (uniform)	1.18 var.	+0.5	+0.1
Automatic stabilizer (targeted)	1.32 var.	+1.5	+0.1
Developing economy (baseline)	2.640	+2.7	+0.4

Welfare effects measured as consumption-equivalent lifetime change relative to the no-policy counterfactual. The stabilizer rows report the output-volatility-reduction ratio rather than an impact multiplier. All experiments use the baseline calibration of Table 2 modified as described in the text.

The four experiments converge on a single take-home: fiscal policy design matters as much as fiscal policy size. Allocating a fixed fiscal envelope toward categories used by constrained households generates larger aggregate multipliers and more progressive welfare effects simultaneously. Cutting those categories has both larger and more regressive contractionary effects. The distributional incidence of public goods is therefore a first-order macroeconomic consideration – not merely a redistribution question.

7 Conclusion

We have shown that Edgeworth complementarity between public and private consumption, combined with empirically motivated heterogeneity in how constrained and unconstrained households internalize publicly provided goods, transforms three features of fiscal transmission. First, a new sufficient statistic – the direct fiscal channel ξ – emerges alongside the standard income-cyclicality statistic χ, giving researchers two independently-identifiable parameters to discipline fiscal multipliers from data. Second, the curvature heterogeneity induced by heterogeneous internalization decouples the aggregate MPC from the aggregate MPE, providing a new route to the MPC/MPE trilemma resolution that operates through the government-spending margin and is orthogonal to existing labor-supply-based resolutions. Third, a distributional dominance condition links inequality, complementarity, and multiplier size: fiscal policy is most effective precisely in the economies where it is most needed – those with high inequality and strong complementarity between public services and the private consumption of the poor.

Quantitatively, the HANK implementation with empirically disciplined θ^j delivers impact multipliers in the range 1.68–1.84, compared to 1.50 in the separable benchmark – an economically significant amplification that accounts for a meaningful share of the dispersion in the empirical fiscal multiplier literature. Policy experiments reveal that targeted public spending on complementary services can raise multipliers by another 24% relative to untargeted spending, while austerity implemented through cuts in public services has aggregate contractionary effects 44% larger than the separable benchmark predicts – with sharply asymmetric distributional costs falling on the bottom of the distribution. In a developing-economy calibration with higher λ and stronger $|\theta^H|$, the impact multiplier rises to 2.64, consistent with the empirical evidence that fiscal policy is more effective in low-income economies.

These findings have four broader implications. First, the fiscal-multiplier literature's tradition of treating fiscal expansions as a single aggregate object misses a first-order margin: the *composition* of fiscal spending across categories with different degrees of complementarity with private consumption. Second, the welfare costs of austerity are systematically under-estimated by models that suppress the c–g complementarity channel, and these costs fall disproportionately on constrained households. Third, automatic stabilizers that maintain public-service provision during downturns do "double duty": they stabilize aggregate demand *and* protect the private-consumption complementarity of the most vulnerable. Fourth, the framework provides a micro-founded explanation for the reduced-form finding that fiscal multipliers are larger in developing economies, and motivates scaling up fiscal expansion through public services as a poverty-reduction tool.

Several directions for future work emerge. *Empirically*, the household-level identification strategy of Section 5 can be extended to additional public-good categories and to the relationship between θ^j and non-homothetic preferences emphasized by Barthel and Francois [2025]. *Theoretically*, combining the c–g non-separability of this paper with the c–n non-separability of Bilbiie

et al. [2025] would yield a framework with maximum flexibility to match all three MPC/MPE facts simultaneously; we leave the cross-product analysis to future work. *Quantitatively*, the model can be extended to study transmission of monetary policy under heterogeneous θ^j, the interaction between fiscal transmission and monetary normalization at the effective lower bound, and the distributional consequences of fiscal consolidation in European and emerging-market settings.

Most fundamentally, our results suggest that the integration of public goods into household preferences – long recognized as important but rarely operationalized in macroeconomic models – is a productive avenue for macroeconomic research. When households care about publicly provided goods differently depending on their income, the distributional incidence of public spending becomes a first-order macroeconomic variable rather than a distributional side-effect. This is a reframing that aligns the macroeconomic analysis of fiscal policy with the growing evidence from development economics, public finance, and inequality research that public-good provision has shaped the distribution of economic outcomes at the global level.

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A Full Model: Chronological Step-by-Step Derivation

This appendix provides a complete, self-contained derivation of the fiscal multiplier from first principles. We build the model chronologically – preferences, optimality conditions, market clearing, and then the multiplier – working with the general heterogeneous (θ^S, θ^H) case throughout. Every algebraic step is shown explicitly. Special cases are verified at each stage.

A.1 Notation and parameters

For any variable X_t, $\hat{X}_t \equiv (X_t - \bar{X})/\bar{X}$ denotes the log-deviation from steady state, where $\bar{X}$ is the steady-state value.

Parameters.

$\sigma > 0$	Inverse elasticity of intertemporal substitution
$\phi > 0$	Inverse Frisch elasticity of labor supply
$\beta \in (0,1)$	Discount factor
$\lambda \in (0,1)$	Steady-state fraction of hand-to-mouth households
$s \in (0,1)$	Probability saver remains saver (THANK only)
$\tau_D \in (0,1)$	Profit redistribution rate
$\mu \in (0,1)$	Tax progressivity (H pays μ/λ per unit of g)
$\rho_g \in (0,1)$	Persistence of government spending shock

Type-specific complementarity objects. For $j \in \{S, H\}$, under the zero-profit normalization $\bar{c}^H = \bar{c}^S = \bar{c}$:

$$\alpha^j \equiv \frac{\bar{c}}{\bar{c} + \theta^j \bar{g}} \qquad \text{(share of private in effective consumption)}$$

$$\Gamma^j \equiv \frac{\theta^j \bar{g}}{\bar{c} + \theta^j \bar{g}} = 1 - \alpha^j \qquad \text{(complementarity ratio)}$$

$$\psi^j \equiv \frac{\Gamma^j}{\alpha^j} = \frac{\theta^j \bar{g}}{\bar{c}} \qquad \text{(complementarity–wealth ratio).}$$

Note $\alpha^j + \Gamma^j = 1$ always. Under $\theta^j < 0$: $\alpha^j > 1$, $\Gamma^j < 0$. Under $\theta^j > 0$: $\alpha^j < 1$, $\Gamma^j > 0$. Under $\theta^j = 0$: $\alpha^j = 1$, $\Gamma^j = 0$ (Bilbiie [2025]'s separable case).

Derived structural objects.

$$K \equiv \phi + 1 - \phi\tau_D/\lambda \qquad \text{(appears in } H \text{ budget constraint)}$$

$$D \equiv (\phi + \sigma\alpha^H) - K\sigma\lambda(\alpha^H - \alpha^S) \qquad \text{(common denominator)}$$

$$\zeta \equiv \frac{\phi}{\phi + \sigma} \qquad \text{(Bilbiie [2025]'s transfer elasticity, separable)}$$

$$\zeta_\alpha \equiv \frac{\phi}{\phi + \sigma\alpha} \qquad \text{(modified, symmetric } \theta\text{).}$$

A.2 Model primitives

Households and preferences. A fraction λ are hand-to-mouth (H); $1-\lambda$ are savers (S). All have utility over effective consumption $\tilde{c}_t^j = c_t^j + \theta^j g_t$ and labor n_t^j:

$$U^j = \frac{(\tilde{c}_t^j)^{1-\sigma}}{1-\sigma} - \frac{(n_t^j)^{1+\phi}}{1+\phi}.$$

Why $\theta^j < 0$ implies Edgeworth complementarity. The cross-partial:

$$U_{cg}^j = \frac{\partial^2 U^j}{\partial c^j \partial g} = -\sigma(\tilde{c}^j)^{-\sigma-1} \cdot \theta^j.$$

Since $\sigma > 0$ and $\tilde{c}^j > 0$: $U_{cg}^j > 0$ iff $\theta^j < 0$. A rise in g raises the marginal utility of c^j – definition of Edgeworth complementarity.

Linearizing effective consumption. $\tilde{c}_t^j = c_t^j + \theta^j g_t$. Log-linearizing around steady state: $\bar{\tilde{c}}^j \hat{\tilde{c}}_t^j = \bar{c}\,\hat{c}_t^j + \theta^j \bar{g}\,\hat{g}_t$. Dividing by $\bar{\tilde{c}}^j = \bar{c} + \theta^j \bar{g}$:

$$\hat{\tilde{c}}_t^j = \underbrace{\frac{\bar{c}}{\bar{c} + \theta^j \bar{g}}}_{\alpha^j} \hat{c}_t^j + \underbrace{\frac{\theta^j \bar{g}}{\bar{c} + \theta^j \bar{g}}}_{\Gamma^j} \hat{g}_t = \alpha^j \hat{c}_t^j + \Gamma^j \hat{g}_t. \tag{36}$$

Production and profits. Linear production: $y_t = n_t$. Optimal sales subsidy ensures zero steady-state profits: $\bar{D} = 0$. This gives $\bar{c}^H = \bar{c}^S = \bar{c}$. Log-linearized profits: $\hat{d}_t = -\hat{w}_t$.

Government and balanced budget. $\hat{g}_t = \rho_g \hat{g}_{t-1} + \varepsilon_t^g$. Balanced budget $T_t = G_t$ gives $\hat{t}_t = \hat{g}_t$. hand-to-mouth tax: $t_t^H = (\mu/\lambda) g_t$, so $\hat{t}_t^H = (\mu/\lambda)\hat{g}_t$.

A.3 Intratemporal optimality conditions

Both types optimize over labor given their wage:

$$(n_t^j)^\phi = w_t (\tilde{c}_t^j)^{-\sigma}. \tag{37}$$

Log-linearizing: $\phi\hat{n}_t^j = \hat{w}_t - \sigma\hat{\tilde{c}}_t^j$. Substituting (36):

$$\phi\hat{n}_t^j = \hat{w}_t - \sigma(\alpha^j\hat{c}_t^j + \Gamma^j\hat{g}_t), \quad j \in \{S, H\}. \tag{38}$$

A.4 Real wage derivation

Aggregating (38) using $\hat{n}_t = \lambda\hat{n}_t^H + (1-\lambda)\hat{n}_t^S$:

$$\phi\hat{n}_t = \hat{w}_t - \sigma[\lambda\alpha^H\hat{c}_t^H + (1-\lambda)\alpha^S\hat{c}_t^S] - \sigma[\lambda\Gamma^H + (1-\lambda)\Gamma^S]\hat{g}_t. \tag{39}$$

Using $\hat{c}_t^S = (\hat{c}_t - \lambda\hat{c}_t^H)/(1-\lambda)$:

$$\lambda\alpha^H\hat{c}_t^H + (1-\lambda)\alpha^S\hat{c}_t^S = \alpha^S\hat{c}_t + \lambda(\alpha^H - \alpha^S)\hat{c}_t^H.$$

Using $\hat{n}_t = \hat{y}_t = \hat{c}_t + \hat{g}_t$ and solving:

$$\hat{w}_t = (\phi + \sigma\alpha^S)\hat{c}_t + \sigma\lambda(\alpha^H - \alpha^S)\hat{c}_t^H + [\phi + \sigma\lambda\Gamma^H + \sigma(1-\lambda)\Gamma^S]\hat{g}_t. \tag{40}$$

Conjecturing $\hat{c}_t^H = \chi\hat{c}_t + \xi\hat{g}_t$ and substituting: $\hat{w}_t = W_c\hat{c}_t + W_g\hat{g}_t$ where $W_c = (\phi + \sigma\alpha^S) + \sigma\lambda\chi(\alpha^H - \alpha^S)$ and $W_g = \phi + \sigma\lambda\Gamma^H + \sigma(1-\lambda)\Gamma^S + \sigma\lambda\xi(\alpha^H - \alpha^S)$.

A.5 Hand-to-mouth budget constraint

hand-to-mouth consume disposable income. Using $\hat{d}_t = -\hat{w}_t$ and $\hat{t}_t^H = (\mu/\lambda)\hat{g}_t$:

$$\hat{c}_t^H = \hat{w}_t\Big(1 - \frac{\tau_D}{\lambda}\Big) + \hat{n}_t^H - \frac{\mu}{\lambda}\hat{g}_t. \tag{41}$$

Substituting $\hat{n}_t^H = (1/\phi)[\hat{w}_t - \sigma\alpha^H\hat{c}_t^H - \sigma\Gamma^H\hat{g}_t]$:

$$\hat{c}_t^H = \hat{w}_t\Big(1 - \frac{\tau_D}{\lambda}\Big) + \frac{\hat{w}_t}{\phi} - \frac{\sigma\alpha^H}{\phi}\hat{c}_t^H - \frac{\sigma\Gamma^H}{\phi}\hat{g}_t - \frac{\mu}{\lambda}\hat{g}_t.$$

Collecting $\hat{c}_t^H$ and multiplying by ϕ:

$$(\phi + \sigma\alpha^H)\hat{c}_t^H = K\hat{w}_t - \Big(\sigma\Gamma^H + \frac{\mu\phi}{\lambda}\Big)\hat{g}_t. \tag{42}$$

A.6 Solving for χ and ξ

Substituting the real wage (40) and conjectured $\hat{c}_t^H = \chi\hat{c}_t + \xi\hat{g}_t$ into (42):

$$(\phi + \sigma\alpha^H)(\chi\hat{c}_t + \xi\hat{g}_t) = K(W_c\hat{c}_t + W_g\hat{g}_t) - (\sigma\Gamma^H + \mu\phi/\lambda)\hat{g}_t.$$

Matching $\hat{c}_t$ coefficients:

$$(\phi + \sigma\alpha^H)\chi = K[(\phi + \sigma\alpha^S) + \sigma\lambda\chi(\alpha^H - \alpha^S)],$$

which gives

$$\chi\big[(\phi + \sigma\alpha^H) - K\sigma\lambda(\alpha^H - \alpha^S)\big] = K(\phi + \sigma\alpha^S),$$

so

$$\chi(\theta^S, \theta^H) = \frac{K(\phi + \sigma\alpha^S)}{D}.$$

Verification 1 (symmetric $\alpha^H = \alpha^S = \alpha$): $D = \phi + \sigma\alpha$, $\chi = K$. ✓

Verification 2 (separable $\alpha^j = 1$): $D = \phi + \sigma$, $\chi = K$. ✓

Matching $\hat{g}_t$ coefficients:

$$(\phi + \sigma\alpha^H)\xi = KW_g - \sigma\Gamma^H - \mu\phi/\lambda,$$

which gives

$$\xi\, D = K[\phi + \sigma\lambda\Gamma^H + \sigma(1-\lambda)\Gamma^S] - \sigma\Gamma^H - \mu\phi/\lambda.$$

Verification 1 (separable): $\xi = \zeta(\chi - \mu/\lambda)$. ✓

Verification 2 (symmetric): $\xi = \zeta_\alpha\big[\chi - \mu/\lambda + \sigma\Gamma(\chi - 1)/\phi\big]$. ✓

A.7 Saver Euler equation

The saver Euler equation: $(\tilde{c}_t^S)^{-\sigma} = \beta R_t \mathbb{E}_t(\tilde{c}_{t+1}^S)^{-\sigma}$. Log-linearizing: $-\sigma(\alpha^S\hat{c}_t^S + \Gamma^S\hat{g}_t) = \hat{r}_t - \sigma\mathbb{E}_t(\alpha^S\hat{c}_{t+1}^S + \Gamma^S\hat{g}_{t+1})$. Dividing by $-\sigma\alpha^S$:

$$\hat{c}_t^S = \mathbb{E}_t\hat{c}_{t+1}^S - \frac{1}{\sigma\alpha^S}\hat{r}_t - \frac{\Gamma^S}{\alpha^S}(\hat{g}_t - \mathbb{E}_t\hat{g}_{t+1}).$$

A.8 THANK self-insurance Euler equation

In THANK, savers face risk of becoming hand-to-mouth:

$$(\tilde{c}_t^S)^{-\sigma} = \beta R_t \mathbb{E}_t[s(\tilde{c}_{t+1}^S)^{-\sigma} + (1-s)(\tilde{c}_{t+1}^H)^{-\sigma}]. \tag{43}$$

Log-linearizing, using $\bar{U}_c^H/\bar{U}_c^S = (\alpha^H/\alpha^S)^\sigma$: under symmetric preferences ($\alpha^H = \alpha^S = \alpha$):

$$\hat{c}_t^S = s\mathbb{E}_t\hat{c}_{t+1}^S + (1-s)\mathbb{E}_t\hat{c}_{t+1}^H - \frac{1}{\sigma\alpha}\hat{r}_t - \frac{\Gamma}{\alpha}(\hat{g}_t - \mathbb{E}_t\hat{g}_{t+1}).$$

A.9 Planned expenditure curve and on-impact multiplier

Aggregate demand closes the model. Substituting $\hat{c}_t = \lambda \hat{c}_t^H + (1-\lambda)\hat{c}_t^S$ into the Euler equation (under symmetric α) and using $\hat{y}_t = \hat{c}_t + \hat{g}_t$:

$$\hat{c}_t = [1 - \beta(1-\lambda\chi)]\hat{y}_t - \frac{(1-\lambda)\beta}{\sigma\alpha}\hat{r}_t + \beta\delta(1-\lambda\chi)\mathbb{E}_t\hat{c}_{t+1} + \beta\lambda\xi(\hat{g}_t - \mathbb{E}_t\hat{g}_{t+1}), \tag{44}$$

where $\delta \equiv 1 + (\chi - 1)(1-s)/(1-\lambda\chi)$.

Under $\hat{r}_t = 0$, $\rho_g = 0$ (transitory shock), the on-impact multiplier follows from:

$$(1-\lambda\chi)\hat{y}_t = (1-\lambda\chi+\lambda\xi)\hat{g}_t \implies \mu = 1 + \frac{\lambda\xi}{1-\lambda\chi}.$$

A.10 Special cases summary

- Separable ($\theta^j = 0$, $\alpha^j = 1$, $\Gamma^j = 0$): $\xi = \zeta(\chi - \mu/\lambda)$, recovering Bilbiie [2025].
- Symmetric ($\theta^S = \theta^H = \theta$, $\alpha^H = \alpha^S = \alpha$): $\xi = \zeta_\alpha[\chi - \mu/\lambda + \sigma\Gamma(\chi-1)/\phi]$.
- RANK limit ($\lambda \to 0$): $\mu = 1 - \psi^S$, saver direct-demand only.
- Heterogeneous ($\theta^H \neq \theta^S$): general χ, ξ expressions with curvature-induced decoupling of MPC and MPE.

B Alternative Derivations of χ and ξ

This appendix provides three alternative derivations of the key sufficient statistics χ and ξ that yield the same expressions by different routes. The methods illuminate different aspects of the model structure and are useful for researchers who wish to extend the framework.

B.1 Why three methods give the same answer

The conjecture $\hat{c}_t^H = \chi\hat{c}_t + \xi\hat{g}_t$ is the unique solution to a system of three equilibrium conditions: the hand-to-mouth budget constraint, the real-wage aggregation, and the production function $\hat{y}_t = \hat{c}_t + \hat{g}_t$. Any method that correctly exploits this system must recover the same (χ, ξ). The three methods differ in how they organize the algebra: (1) *Coefficient matching*: used in Appendix A. (2) *Matrix system*: expresses the equilibrium conditions as a linear system and solves by inversion. (3) *Direct substitution*: substitutes the wage directly into the hand-to-mouth budget constraint and solves for $\hat{c}_t^H$ as a function of $\hat{c}_t$ and $\hat{g}_t$.

B.2 Method 1: coefficient matching (main text)

Reproduced in Appendix A for completeness. The key algebraic step is matching the coefficients of $\hat{c}_t$ and $\hat{g}_t$ in the equation $(\phi + \sigma\alpha^H)\hat{c}_t^H = KW_c\hat{c}_t + [KW_g - \sigma\Gamma^H - \mu\phi/\lambda]\hat{g}_t$.

B.3 Method 2: matrix system

Write the system as $A\mathbf{x} = B\mathbf{g}$ where $\mathbf{x} = (\hat{c}^H, \hat{c}^S, \hat{w}, \hat{n})'$ and $\mathbf{g} = (\hat{c}_t, \hat{g}_t)'$:

$$A = \begin{pmatrix} \phi + \sigma\alpha^H & 0 & -K & 0 \\ 0 & \phi + \sigma\alpha^S & -1 & 0 \\ 0 & 0 & 1 & -\phi \\ -\lambda & -(1-\lambda) & 0 & 1 \end{pmatrix},$$

and solve for $\hat{c}^H = f(\hat{c}, \hat{g})$. The χ and ξ expressions match those in the main text.

B.4 Method 3: direct substitution

The hand-to-mouth consumption $\hat{c}_t^H$ is determined by combining the hand-to-mouth budget constraint with the real-wage equation. Direct substitution involves: (i) solving the wage aggregation for $\hat{w}_t$ as a function of $\hat{c}_t, \hat{g}_t, \hat{c}_t^H$; (ii) substituting into the hand-to-mouth budget (42); (iii) solving for $\hat{c}_t^H$. The result coincides with Methods 1 and 2.

B.5 What χ really is: income cyclicality

Having derived χ three ways, it is worth emphasizing the economic interpretation. χ is the elasticity of hand-to-mouth consumption with respect to aggregate output, holding government spending fixed: $\chi = \partial\hat{c}^H/\partial\hat{y}\mid_{\hat{g}=0}$. Since hand-to-mouth consume their income, χ is also the elasticity of hand-to-mouth income to aggregate output – the "income cyclicality" terminology. Under countercyclical inequality ($\chi > 1$), hand-to-mouth income rises more than proportionally with aggregate output, amplifying the fiscal multiplier through the NK Cross.

C Derivation Under General Non-Separable Preferences

This appendix provides the full derivation of χ and ξ under Assumption 1, working with the general utility elasticities $(\epsilon_{cc}^j, \epsilon_{cg}^j)$ rather than the linear effective consumption parameterization.

C.1 Linearizing the intratemporal condition

The intratemporal FOC $-U_n^j(c^j, g, n^j) = w_t U_c^j(c^j, g, n^j)$. Log-linearizing around steady state, using $U_{cc}^j = -(\bar{U}_c^j/\bar{c})\epsilon_{cc}^j$ and $U_{cg}^j = -(\bar{U}_c^j/\bar{g})\epsilon_{cg}^j$:

$$\phi\hat{n}_t^j = \hat{w}_t - \epsilon_{cc}^j\hat{c}_t^j - \epsilon_{cg}^j\hat{g}_t. \tag{45}$$

Under Assumption 2, $\epsilon_{cc}^j = \sigma\alpha^j$ and $\epsilon_{cg}^j = \sigma\Gamma^j$.

C.2 Linearizing the saver Euler equation

The Euler equation $U_c^S(c_t^S, g_t) = \beta R_t \mathbb{E}_t U_c^S(c_{t+1}^S, g_{t+1})$. Log-linearizing:

$$-\epsilon_{cc}^S \hat{c}_t^S - \epsilon_{cg}^S \hat{g}_t = \hat{r}_t - \mathbb{E}_t[\epsilon_{cc}^S \hat{c}_{t+1}^S + \epsilon_{cg}^S \hat{g}_{t+1}].$$

Dividing by $-\epsilon_{cc}^S$:

$$\hat{c}_t^S = \mathbb{E}_t \hat{c}_{t+1}^S - \frac{1}{\epsilon_{cc}^S}\hat{r}_t - \frac{\epsilon_{cg}^S}{\epsilon_{cc}^S}(\hat{g}_t - \mathbb{E}_t \hat{g}_{t+1}). \tag{46}$$

C.3 Verification: linear effective consumption

Under Assumption 2 with $\epsilon_{cc}^S = \sigma\alpha^S$ and $\epsilon_{cg}^S = \sigma\Gamma^S$:

$$\hat{c}_t^S = \mathbb{E}_t \hat{c}_{t+1}^S - \frac{1}{\sigma\alpha^S}\hat{r}_t - \frac{\Gamma^S}{\alpha^S}(\hat{g}_t - \mathbb{E}_t \hat{g}_{t+1}),$$

which matches (24).

C.4 General χ and ξ: full derivation

Starting from (45) for type H and the H budget constraint $\hat{c}_t^H = \hat{w}_t(1 - \tau_D/\lambda) + \hat{n}_t^H - \hat{t}_t^H$, substituting the labor FOC and solving:

$$(\phi + \epsilon_{cc}^H)\hat{c}_t^H = K\hat{w}_t - (\epsilon_{cg}^H + \mu\phi/\lambda)\hat{g}_t. \tag{47}$$

The aggregated wage equation is

$$\hat{w}_t = (\phi + \epsilon_{cc}^S)\hat{c}_t + \lambda\chi(\epsilon_{cc}^H - \epsilon_{cc}^S)\hat{c}_t + \lambda(\epsilon_{cg}^H + \epsilon_{cg}^S)\hat{g}_t + \cdots \tag{48}$$

(details analogous to Appendix A.4). Matching coefficients:

$$\chi = \frac{K(\phi + \epsilon_{cc}^S)}{D}, \qquad D = (\phi + \epsilon_{cc}^H) - K\lambda(\epsilon_{cc}^H - \epsilon_{cc}^S),$$

$$\xi = \frac{K[\phi + \lambda\epsilon_{cg}^H + (1-\lambda)\epsilon_{cg}^S] - \epsilon_{cg}^H - \mu\phi/\lambda}{D}.$$

C.5 The separability of χ from ϵ_{cg}^j

A key feature of the general expression is that χ depends only on ϵ_{cc}^j, *not* on ϵ_{cg}^j. Intuitively, income cyclicality measures how hand-to-mouth income co-moves with aggregate output, which is determined by the labor-supply-wage response governed by own-consumption curvature. Non-separability between c and g enters only through the direct fiscal channel ξ, which is determined

by cross-elasticities. This separation is a direct consequence of linearization and is a general property of the class of non-separable preferences considered.

D Full Model Setup and Optimization Problems

This appendix provides the complete model setup, including primitive optimization problems for each agent, market clearing conditions, equilibrium definition, and the mapping from first-order conditions to the log-linearized equilibrium system.

D.1 Environment

Time is discrete, $t = 0, 1, 2, \ldots$. The economy has three sectors: households (continuum), firms (continuum), and government. All shocks are realized at the start of the period.

D.2 Households

Household i is either a saver or hand-to-mouth in period t, with type switching stochastically. Each household maximizes $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U^j(c_{it}^j, g_t, n_{it}^j)$ subject to type-specific constraints.

Saver problem.

$$\begin{aligned}
&\max_{\{c_{it}^S, n_{it}^S, b_{it}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U^S(\tilde{c}_{it}^S, n_{it}^S) \\
&\quad \text{s.t.} \quad c_{it}^S + b_{it} = w_t n_{it}^S + R_{t-1} b_{it-1} + T_{it}^S + \tau_D D_{it}^S, \\
&\qquad\quad\; b_{it} \geq \underline{b}, \qquad \tilde{c}_{it}^S = c_{it}^S + \theta^S g_t.
\end{aligned}$$

hand-to-mouth problem.

$$\begin{aligned}
&\max_{\{c_{it}^H, n_{it}^H\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U^H(\tilde{c}_{it}^H, n_{it}^H) \\
&\quad \text{s.t.} \quad c_{it}^H = w_t n_{it}^H + T_{it}^H - t_{it}^H, \qquad \tilde{c}_{it}^H = c_{it}^H + \theta^H g_t.
\end{aligned}$$

D.3 Final goods firm

A competitive final-goods firm assembles intermediate goods $y_t(j)$ using $Y_t = \left[\int_0^1 y_t(j)^{(\varepsilon-1)/\varepsilon} dj\right]^{\varepsilon/(\varepsilon-1)}$. Cost minimization gives demand for each intermediate: $y_t(j) = (p_t(j)/P_t)^{-\varepsilon} Y_t$.

D.4 Intermediate goods firms and Rotemberg pricing

Intermediate firm j produces $y_t(j) = n_t(j)$ with marginal cost w_t. Under Rotemberg adjustment costs, firm j maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \Lambda_t \Big[\frac{p_t(j)}{P_t} y_t(j) - w_t n_t(j) - \frac{\psi_p}{2} \Big(\frac{p_t(j)}{p_{t-1}(j)} - 1 \Big)^2 Y_t \Big],$$

where Λ_t is the stochastic discount factor of the savers. The first-order condition yields the Rotemberg Phillips curve (11).

D.5 Government and monetary policy

Government budget constraint:

$$G_t + R_{t-1} B_{t-1} = T_t + B_t,$$

with balanced budget $T_t = G_t$ for analytical results. Central bank follows Taylor rule $\hat{\iota}_t = \phi_\pi \pi_t + \phi_y \hat{y}_t$.

D.6 Market clearing

Labor: $N_t = \lambda n_t^H + (1-\lambda) n_t^S$. Goods: $Y_t = C_t + G_t$. Bonds: $B_t = 0$ in equilibrium. Consumption aggregate: $C_t = \lambda c_t^H + (1-\lambda) c_t^S$.

D.7 Equilibrium definition

A competitive equilibrium consists of sequences of allocations, prices, and government variables such that: (i) households optimize given prices; (ii) firms optimize given wages and aggregate demand; (iii) government budget is satisfied; (iv) markets clear.

D.8 From FOCs to equilibrium conditions

Log-linearizing the equilibrium conditions around the symmetric zero-inflation steady state gives the system (4), (5), the Phillips curve (11), the Taylor rule (14), and market clearing (15). The analytical results of Section 3 follow from this system by imposing $\hat{r}_t = 0$ and solving for the on-impact multiplier.

E Calibration, Simulation Code, and Figures

This appendix documents the calibration choices, provides key multiplier values under alternative parameterizations, and describes the simulation code used to generate all figures in the paper.

E.1 Baseline calibration

The baseline calibration is reported in Table 2 in the main text. The steady-state government spending share $\bar{g}/\bar{y} = 0.20$ implies $\bar{g}/\bar{c} = 0.25$, used in $\alpha^j = \bar{c}/(\bar{c} + \theta^j \bar{g})$.

E.2 Key multiplier values under alternative calibrations

Table 8: Key multiplier values across alternative calibrations.

Case	χ	ξ	Multiplier μ
Bilbiie separable ($\theta = 0$, $\lambda = 0.35$)	2.000	0.429	1.500
Symmetric complementarity ($\theta = -0.5$, $\lambda = 0.35$)	2.000	0.304	1.355
Symmetric substitutability ($\theta = +0.3$, $\lambda = 0.35$)	2.000	0.498	1.581
Heterogeneous internalization ($\theta^S = +0.29, \theta^H = -0.76, \lambda = 0.35$)	1.874	0.530	1.539
Estimated upper bound ($\theta^S = +0.38, \theta^H = -0.90$)	1.802	0.611	1.612
Developing economy ($\lambda = 0.70, \theta^H = -1.20$)	2.345	0.890	2.640
Determinacy boundary ($\lambda\chi = 1$)	—	—	∞

Note on the determinacy boundary: at $\lambda = 0.50$ with $\theta = -0.5$, $\chi = 2.0$ exactly, so $\lambda\chi = 1$ and the multiplier diverges. This is the well-known HANK indeterminacy region identified by Bilbiie [2008]: when $\lambda\chi \geq 1$, the NK Cross feedback loop is explosive and the equilibrium is indeterminate. All simulations use $\lambda = 0.35$ or $\lambda = 0.70$, both of which are safely below the boundary.

E.3 Figures

All figures in the paper are generated by the Python simulation code in `code/simulation/simulations.py` and `code/hank/hank_jacobian.py`. The code implements all analytical functions $(\alpha^j, \Gamma^j, \chi, \xi, \mu)$ directly from the closed-form expressions in Section 3, verifies the multiplier ranking of Corollaries 4 and 5 numerically, solves the HANK model via the sequence-space Jacobian method, and generates all figures referenced in Sections 4 and 6.

The full code is open-source and available in the replication repository (see README.md). Dependencies: `numpy`, `matplotlib`, `scipy`, `pandas`.

E.4 Simulation code

The simulation is organized into four modules:

- `code/simulation/analytical.py`: computes χ, ξ, μ from the closed-form expressions.
- `code/simulation/figures.py`: generates all paper figures.
- `code/hank/hank_jacobian.py`: solves the HANK model via the sequence-space Jacobian method.
- `code/empirical/cex_analysis.do`: implements the CEX diff-in-diff estimation of Section 5.

Run `python3 run_all.py` from the repository root to reproduce all quantitative results and figures in the paper.